HAIMAN'S CONJECTURE FOR $n \leq 10$: KAZHDAN–LUSZTIG POSITIVITY OF MONOMIAL IMMANANTS

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Abstract. Haiman conjectured that for every partition $\lambda \vdash n$ and every $w \in S_n$ the polynomial $c_{\lambda,w}(q)$, a monomial character of the Hecke algebra evaluated at the Kazhdan–Lusztig basis, has non-negative integer coefficients and is symmetric and unimodal, and he verified this by computer for $n \leq 7$. We verify it for every $n \leq 10$. A stability lemma reduces the statement to $n = 10$, which we settle by an exact computation that stores only the 1,252,173,030 pairs (x, w) in S_{10} with x maximal in its double coset for the descent sets of w, and takes the traces of the standard basis from the class polynomials of Geck and Pfeiffer. Every stored Kazhdan–Lusztig polynomial agrees with du Cloux's Coxeter3, and the coefficients pass several independent checks, including a comparison at all 16,796 codominant w with chromatic quasisymmetric functions computed from colourings.

Consequently, for $n \leq 10$ every monomial immanant is a non-negative integer combination of Kazhdan–Lusztig immanants, Stembridge's monomial immanant conjecture holds, and a class-function immanant is non-negative on totally non-negative matrices if and only if it is non-negative at the block matrices J_μ. We use this to describe the order of normalised irreducible-character immanants on totally non-negative matrices for $n \leq 10$, and to prove Pate's inequalities there. On the full span of permutation monomials we give a Newton-polytope criterion for the coefficient of Imm_w to be non-negative throughout the totally non-negative cone, show that for $n = 4, 5$ this happens exactly when X_w is smooth, and check by exact computation for $n \leq 10$ the proof of this for all n given in a companion paper.

1. Introduction

A real matrix is *totally non-negative* (TNN) if all its minors are non-negative; TNN_n denotes the set of such $n \times n$ matrices. For $f\colon S_n \to \mathbb{R}$ and an $n \times n$ matrix x put $d_f(x) = \sum_\sigma f(\sigma)\, x_{1,\sigma(1)} \cdots x_{n,\sigma(n)}$. For an irreducible character χ^λ this is the immanant $\mathrm{Imm}_{\chi^\lambda}$, and Stembridge [24] proved that it is non-negative on TNN_n. The monomial immanants $\mathrm{Imm}_\lambda = d_{\phi^\lambda}$, where ϕ^λ is the monomial virtual character, refine the irreducible ones, and Stembridge conjectured [25, Conj. 2.1] that they are non-negative on TNN_n too. Haiman [9] proposed a stronger statement in the Hecke algebra. Let H_n be the Hecke algebra of S_n over $\mathbb{Z}[q^{\pm 1/2}]$, with $(T_s - q)(T_s + 1) = 0$, let $P_{v,w}(q)$ be the Kazhdan–Lusztig polynomials, so that $q^{\ell(w)/2} C'_w = \sum_{v \leq w} P_{v,w}(q)\, T_v$, and let ϕ^λ_q be the monomial characters of H_n (Section 2.3).

Conjecture 1.1 (Haiman [9, Conj. 2.1])**.** *For every $\lambda \vdash n$ and $w \in S_n$, the polynomial $c_{\lambda,w}(q) = \phi^\lambda_q(q^{\ell(w)/2} C'_w)$ has non-negative integer coefficients, and it is symmetric and unimodal about $q^{\ell(w)/2}$.*

At $q = 1$, C'_w specialises to $C'_w(1) = \sum_{v \leq w} P_{v,w}(1)\, v$, and Conjecture 1.1 gives the following.

Conjecture 1.2 (Haiman, Stembridge; $q = 1$)**.** *For every $\lambda \vdash n$ and $w \in S_n$, $c_{\lambda,w} = c_{\lambda,w}(1) = \phi^\lambda(C'_w(1)) \geq 0$.*

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Rhoades and Skandera [19] showed that $\mathrm{Imm}_\lambda = \sum_w c_{\lambda,w} \mathrm{Imm}_w$, where the Kazhdan–Lusztig immanants Imm_w are non-negative on TNN_n. So Conjecture 1.2 implies Stembridge's conjecture. It is known for shapes with at most two columns [3, 19], and for smooth w and every shape [2, 1, 10], [22, Prop. 25]; the smooth case depends on Hikita's theorem [10]. Haiman verified Conjecture 1.1 by computer for $n \leq 7$ [9, §4]; earlier, Stembridge had checked his conjecture for $n \leq 4$ and for $n = 5$ apart from two shapes [25]. The companion preprint [13] gives a proof for all hooks and every n. Non-negativity of hook monomial immanants on TNN_n, Stembridge's conjecture for hooks, was proved earlier by Dong, Wu and Gao [5], who need, for the hook $(n-k, 1^k)$, only the minors of order at most $k+1$ to be non-negative. Our main result extends Haiman's verification to $n \leq 10$.

Theorem 1.3. *For every $n \leq 10$, every $\lambda \vdash n$ and every $w \in S_n$, $c_{\lambda,w}(q)$ has non-negative integer coefficients and is symmetric and unimodal about $q^{\ell(w)/2}$. In particular $c_{\lambda,w} = c_{\lambda,w}(1)$ is a non-negative integer.*

For codominant (312-avoiding) w, $c_{\lambda,w}(q)$ is the coefficient of e_λ in the chromatic quasisymmetric function of the indifference graph of w (see [2, 20]; for $n \leq 10$ we check this identity directly, Section 7, check 13), and Conjecture 1.1 at these w is the conjecture of Shareshian and Wachs that these functions are e-positive and e-unimodal [20, Conj. 5.1]. At $q = 1$, e-positivity is the Stanley–Stembridge conjecture for natural unit interval orders, proved by Hikita [10]; the e-positivity of the individual coefficients of q^j, and e-unimodality, remain open in general. Shareshian and Wachs checked their conjecture by computer for the graphs $G_{n,r}$ with $n \leq 8$ [20, §5], and through [2] Haiman's computation [9, §4] covers all natural unit interval orders with at most 7 elements. Theorem 1.3 covers all those with at most 10 elements; we are not aware of a published check of all natural unit interval orders with 8 or more elements.

Three cones. We state the consequences in terms of three cones in the space CF_n of real class functions on S_n:

$$\begin{aligned} C_{\mathrm{KL}} &= \{\phi : \ \phi(C'_w(1)) \geq 0 \text{ for all } w \in S_n\}, \\ C_{\mathrm{TNN}} &= \{\phi : \ d_\phi \geq 0 \text{ on } \mathrm{TNN}_n\}, \\ C_J &= \{\phi : \ d_\phi(J_\mu) \geq 0 \text{ for all } \mu \vdash n\}, \end{aligned}$$

where J_μ is the block-diagonal matrix with all-ones blocks of sizes $\mu_1, \mu_2, \ldots$. For every n, $C_{\mathrm{KL}} \subseteq C_{\mathrm{TNN}} \subseteq C_J$, and C_J is the simplicial cone spanned by the monomial characters (Section 3). So Stembridge's conjecture is the equality $C_{\mathrm{TNN}} = C_J$, and Conjecture 1.2 is $C_{\mathrm{KL}} = C_J$. Theorem 1.3 gives both for $n \leq 10$.

Corollary 1.4. *For $n \leq 10$, $C_{\mathrm{KL}} = C_{\mathrm{TNN}} = C_J$. Hence every monomial immanant of size $n \leq 10$ is non-negative on TNN_n, and a class-function immanant d_ϕ is non-negative on TNN_n iff $d_\phi(J_\mu) \geq 0$ for all $\mu \vdash n$, iff it is a non-negative combination of Kazhdan–Lusztig immanants.*

For the normalised immanants $N_a = \mathrm{Imm}_{\chi^a} / \chi^a(e)$, Corollary 1.4 turns every inequality $N_a \geq N_b$ on TNN_n into a finite check on Kostka numbers $K_{a,\mu}$.

Theorem 1.5. *Let $n \leq 10$ and $a, b \vdash n$, and write $f^a = \chi^a(e)$. Then $N_a \geq N_b$ on TNN_n iff $K_{a,\mu}/f^a \geq K_{b,\mu}/f^b$ for every $\mu \vdash n$. If the criterion fails at μ, then $N_a(J_\mu) < N_b(J_\mu)$.*

For $a, b \vdash n$ write $a \succeq b$ if $N_a \geq N_b$ on TNN_n. This is a partial order on the partitions of n for every n, and it is contained in dominance order, since $N_a(J_b) \geq N_b(J_b) > 0$ forces $K_{a,b} > 0$ (Section 8). For $n \leq 10$ it equals dominance only when $n \leq 3$, and it is a lattice only when $n \leq 5$ (Theorem 8.2); we make no claim about its structure for $n \geq 11$. We also obtain Pate's inequalities on TNN_n for $n \leq 10$ (Corollary 8.3), answering Skandera's question [22, Problem 29]

in this range, and an inequality between several normalised immanants that is not a consequence of two-term inequalities (Proposition 8.5).

The computation. Lemma 4.1 shows that $c_{\lambda,w}$ is unchanged when w is extended by fixed points and λ by parts equal to 1, so Theorem 1.3 for $n = 10$ implies it for all $n \le 10$. A direct computation of all Kazhdan–Lusztig polynomials of S_{10} is out of reach: S_9 already has 10,501,351,657 Bruhat pairs, and S_{10} about 10^{12}. By descent invariance it suffices to store $P_{x,w}$ for the x that are maximal in their double coset for the descent sets of w. In S_{10} there are 1,252,173,030 such *extremal pairs*. Working with extremal pairs is not new: du Cloux's Coxeter3 [4] uses them, and Warrington's tables of Kazhdan–Lusztig polynomials store a compressed set of them for $S_4, \dots, S_9$ [26]. Our contributions are their enumeration as contingency tables (Lemma 5.3) and the scale of the computation. We run the Kazhdan–Lusztig recursion on the extremal pairs (Section 6) and sum characters over double cosets (Section 6.1); for the graded statement the characters are replaced by the traces $\phi^\lambda_q(T_v)$, computed from the class polynomials of Geck and Pfeiffer [8] (Section 6.2). At $q = 1$ the computation takes under five minutes on 192 cores, and the graded one an hour and a half. Section 7 lists the checks, including an exhaustive comparison of all 1.25×10^9 extremal pairs with du Cloux's Coxeter3 [4].

Beyond class functions. On the span Π_n of all permutation monomials, the analogous cones $C_{\mathrm{KL}}(\Pi_n) \subseteq C_{\mathrm{TNN}}(\Pi_n)$ differ for $n \ge 4$ [19, 21]. Answering a question of Rhoades and Skandera [19], Skandera [21] showed that whenever the Schubert variety X_w is smooth, the coefficient of Imm_w is non-negative in every element of $C_{\mathrm{TNN}}(\Pi_n)$. We give a vertex criterion on Newton polytopes for this property (Theorem 9.1), and show with it that for $n \le 5$ it holds exactly when X_w is smooth (Theorem 9.2). The companion paper gives a proof of this for every n [13, Thm. 9.6], by a different method, and Section 10 checks that proof for $n \le 10$. So the restriction to class functions in Corollary 1.4 is essential.

The companion paper [13] gives proofs of the results that we state for every n, among them Kazhdan–Lusztig positivity of all hook monomial immanants. Section 10 records the computations that support the conjectures stated there. The paper is organised linearly: Sections 2–4 set up the cones and the reduction to $n = 10$, Sections 5–7 carry out the computation, Section 8 derives the consequences, Section 9 treats the full span Π_n, and Section 10 collects further computations. The code, the verification scripts and the result files are archived as [14], and the complete table of coefficients for $n = 10$ separately as [15]; Appendix B explains how to reproduce and check the computation.

2. Preliminaries

2.1. **Class functions.** For a composition α of n let $S_\alpha = S_{\alpha_1} \times S_{\alpha_2} \times \cdots$ be the Young subgroup preserving the consecutive blocks of sizes $\alpha_1, \alpha_2, \dots$, let $\alpha! = |S_\alpha|$, and let $w_0(\alpha)$ be the longest element of S_α. For $\mu \vdash n$ put $\eta^\mu = \mathrm{Ind}_{S_\mu}^{S_n} 1$. With $\langle \phi, \psi \rangle = \frac{1}{n!} \sum_\sigma \phi(\sigma)\psi(\sigma)$, the monomial characters ϕ^λ are the dual basis: $\langle \phi^\lambda, \eta^\mu \rangle = \delta_{\lambda\mu}$. Under the Frobenius characteristic, $\eta^\mu \mapsto h_\mu$, $\chi^\lambda \mapsto s_\lambda$ and $\phi^\lambda \mapsto m_\lambda$. By Young's rule $\langle \chi^a, \eta^\mu \rangle = K_{a,\mu}$. This is the dimension of the space of S_μ-invariants in the irreducible module V^a, and $K_{a,\mu} \neq 0$ iff a dominates μ.

2.2. **Kazhdan–Lusztig immanants.** Rhoades and Skandera [19] define

$$\mathrm{Imm}_w(x) = \sum_{v \ge w} (-1)^{\ell(v)-\ell(w)} P_{w_0 v, w_0 w}(1)\, x_{1,v(1)} \cdots x_{n,v(n)},$$

and prove that $\mathrm{Imm}_w \ge 0$ on TNN_n and that, for every $f \colon S_n \to \mathbb{R}$,

$$d_f = \sum_{w \in S_n} f(C'_w(1))\ \mathrm{Imm}_w\,. \tag{1}$$

We call d_f *KL-positive* if $f(C'_w(1)) \geq 0$ for all w. Since $P_{v,w} = P_{v^{-1},w^{-1}} = P_{w_0vw_0,w_0ww_0}$, we have $c_{\lambda,w} = c_{\lambda,w^{-1}} = c_{\lambda,w_0ww_0}$.

2.3. **The Hecke algebra.** H_n has basis $\{T_v\}_{v\in S_n}$, with $T_vT_s = T_{vs}$ if $\ell(vs) > \ell(v)$ and $(T_s - q)(T_s + 1) = 0$. For a composition α let H_α be the parabolic subalgebra spanned by the T_v, $v \in S_\alpha$, and let η_q^μ be the character of H_n induced from the index representation $T_s \mapsto q$ of H_μ. The *monomial characters* ϕ_q^λ are defined from the η_q^μ by the same inverse-Kostka relation as at $q = 1$, so that $\chi_q^a = \sum_\lambda K_{a,\lambda}\phi_q^\lambda$ and the Frobenius characteristic of ϕ_q^λ is m_λ; at $q = 1$ they specialise to ϕ^λ. Put

$$c_{\lambda,w}(q) = \phi_q^\lambda\big(q^{\ell(w)/2}C'_w\big) = \sum_{v\leq w} P_{v,w}(q)\,\phi_q^\lambda(T_v),$$

so that $c_{\lambda,w}(1) = c_{\lambda,w}$. Since $P_{v,w} = P_{v^{-1},w^{-1}} = P_{w_0vw_0,w_0ww_0}$ and $\tau(T_v) = \tau(T_{v^{-1}}) = \tau(T_{w_0vw_0})$ for every trace τ of H_n, also $c_{\lambda,w}(q) = c_{\lambda,w^{-1}}(q) = c_{\lambda,w_0ww_0}(q)$. The value $\phi_q^\lambda(T_v)$ is not a function of the conjugacy class of v; Section 6.2 explains how it is computed.

Example 2.A. For $w = 3412$ and $\lambda = (4), (3,1), (2,2), (2,1,1), (1^4)$,

$$c_{\cdot,3412}(q) = \big(1 + 2q + 2q^2 + 2q^3 + q^4,\ q(1+q)^2,\ q(1+q)^2,\ 0,\ 0\big),$$

each symmetric and unimodal about $q^{\ell(w)/2} = q^2$; at $q = 1$ this is $(8,4,4,0,0)$ (Example 6.B).

2.4. **Block reduction.** Say $w = w_1 \times w_2$ if w maps $\{1,\dots,m\}$ to itself, where $w_1 \in S_m$ and $w_2 \in S_{n-m}$ are the restrictions. Every $v \leq w$ then lies in $S_m \times S_{n-m}$, and $P_{v,w} = P_{v_1,w_1}P_{v_2,w_2}$, so $C'_w(1) = C'_{w_1}(1) \otimes C'_{w_2}(1)$ in $\mathbb{Z}[S_m] \otimes \mathbb{Z}[S_{n-m}]$. Restricting ϕ^λ to $S_m \times S_{n-m}$ corresponds to the coproduct $\Delta m_\lambda = \sum_{\mu\cup\nu=\lambda} m_\mu \otimes m_\nu$, where $\mu \cup \nu$ is the partition with the parts of both. This gives:

Lemma 2.1. $c_{\lambda,w_1\times w_2} = \sum_{\mu\cup\nu=\lambda} c_{\mu,w_1}\,c_{\nu,w_2}$.

3. Three cones

Lemma 3.1. (1) *For every composition α of n, $J_\alpha \in \mathrm{TNN}_n$.*

(2) *For every class function ϕ, $d_\phi(J_\mu) = \mu!\,\langle\phi,\eta^\mu\rangle$. In particular $\mathrm{Imm}_\lambda(J_\mu) = \mu!\,\delta_{\lambda\mu}$ and $\mathrm{Imm}_{\chi^a}(J_\mu) = \mu!\,K_{a,\mu}$.*

(3) *For every $w \in S_n$, $\mathrm{Imm}_w(J_\alpha) = \delta_{w,w_0(\alpha)}$. In particular $c_{\lambda,w_0(\alpha)} = \alpha!\,\delta_{\lambda,\mathrm{sort}(\alpha)}$.*

Proof. (1) Order the rows and columns of a square submatrix by the blocks they lie in. If some block contributes more rows than columns, those rows are supported on too few columns and the minor vanishes. Otherwise the minor is a product of determinants of all-ones matrices, each equal to 0 or 1. (2) $\prod_i (J_\mu)_{i,\sigma(i)} = [\sigma \in S_\mu]$, so $d_\phi(J_\mu) = \sum_{\sigma\in S_\mu}\phi(\sigma) = \mu!\,\langle\phi,\eta^\mu\rangle$ by Frobenius reciprocity. (3) The lower interval of $w_0(\alpha)$ is S_α, and $P_{v,w_0(\alpha)} = 1$ for $v \in S_\alpha$, so $C'_{w_0(\alpha)}(1) = \sum_{v\in S_\alpha} v$. For every f, (1) at J_α gives $\sum_w f(C'_w(1))\,\mathrm{Imm}_w(J_\alpha) = \sum_{v\in S_\alpha} f(v) = f(C'_{w_0(\alpha)}(1))$. Since $\{C'_w(1)\}$ is a basis and f is arbitrary, $\mathrm{Imm}_w(J_\alpha) = \delta_{w,w_0(\alpha)}$. The last statement follows from (2). □

Theorem 3.2. *For every n, $C_{\mathrm{KL}} \subseteq C_{\mathrm{TNN}} \subseteq C_J$, and $C_J = \mathrm{cone}\{\phi^\lambda : \lambda \vdash n\}$ is simplicial. Hence Stembridge's conjecture for n is equivalent to $C_{\mathrm{TNN}} = C_J$, and Conjecture 1.2 for n is equivalent to $C_{\mathrm{KL}} = C_J$.*

Proof. The first inclusion is (1) together with $\mathrm{Imm}_w \geq 0$ on TNN_n, and the second is Lemma 3.1(1). By Lemma 3.1(2), $\phi \in C_J$ iff $\langle\phi,\eta^\mu\rangle \geq 0$ for all μ, iff ϕ has non-negative coordinates in the basis $\{\phi^\lambda\}$ dual to $\{\eta^\mu\}$. So $C_{\mathrm{TNN}} = C_J$ iff every generator ϕ^λ lies in C_{TNN}, and similarly for C_{KL}. □

The inclusion $C_{\mathrm{KL}} \subseteq C_J$ also follows directly from Lemma 3.1(3), by evaluation at the elements $w_0(\mu)$. This is due to Haiman, who observed that any class function with the positivity of Conjecture 1.2 is a non-negative combination of monomial characters [9, §2 and Prop. 4.1(1)]. Theorem 3.2 uses no computation and holds for every n. The computation enters the three-cone picture only through the equality $C_{\mathrm{KL}} = C_J$ for $n \le 10$ (Corollary 1.4).

4. Stability

Lemma 4.1 (stability)**.** *Let $\lambda \vdash m$, $w \in S_m$, $k \ge 0$, and let e_k be the identity of S_k. Then $c_{\lambda\cup 1^k,\, w\times e_k} = c_{\lambda,w}$.*

Proof. By Lemma 2.1, $c_{\lambda\cup 1^k, w\times e_k} = \sum_{\mu\cup\nu=\lambda\cup 1^k} c_{\mu,w}\, c_{\nu,e_k}$, with $\nu \vdash k$. Now $C'_{e_k}(1) = e$, and $c_{\nu,e_k} = \phi^\nu(e) = \langle \phi^\nu, \eta^{1^k}\rangle = \delta_{\nu,1^k}$, because η^{1^k} is the regular character. The only term left is $\nu = 1^k$, $\mu = \lambda$. □

Corollary 4.2. *If Conjecture 1.2 holds for N, it holds for every $n \le N$. The same holds for Stembridge's conjecture.*

Proof. For $\lambda \vdash n$ and $w \in S_n$, Lemma 4.1 gives $c_{\lambda,w} = c_{\lambda\cup 1^{N-n},\, w\times e_{N-n}} \ge 0$. For Stembridge's conjecture, let $A \in \mathrm{TNN}_n$. Only $\sigma \in S_n\times\{e\}$ contribute to $d_\phi(A\oplus I_{N-n})$, so $d_\phi(A\oplus I_{N-n}) = d_\psi(A)$ with $\psi(\sigma) = \phi(\sigma \times e)$. For $\phi = \phi^{\lambda\cup 1^{N-n}}$ the coproduct and $\phi^\nu(e) = \delta_{\nu,1^{N-n}}$ give $\psi = \phi^\lambda$, so $\mathrm{Imm}_\lambda(A) = \mathrm{Imm}_{\lambda\cup 1^{N-n}}(A \oplus I) \ge 0$. □

Both statements hold at q.

Lemma 4.3 (graded coproduct and stability)**.** *For $w = w_1 \times w_2$ with $w_1 \in S_m$, $c_{\lambda,w}(q) = \sum_{\mu\cup\nu=\lambda} c_{\mu,w_1}(q)\, c_{\nu,w_2}(q)$. In particular $c_{\lambda\cup 1^k,\, w\times e_k}(q) = c_{\lambda,w}(q)$, and if Conjecture 1.1 holds for N then it holds for every $n \le N$.*

Proof. Since $\ell(w) = \ell(w_1) + \ell(w_2)$ and $P_{v,w} = P_{v_1,w_1}P_{v_2,w_2}$, $q^{\ell(w)/2}C'_w$ is the tensor product of $q^{\ell(w_i)/2}C'_{w_i}$ in $H_m \otimes H_{n-m} = H_{(m,n-m)} \subseteq H_n$. The restriction of η^κ_q to this parabolic subalgebra is given by the Mackey formula for Hecke algebras [8, §9.1]. Explicitly, the double cosets $S_{(m,n-m)}\backslash S_n / S_\kappa$ correspond to the $2\times\ell(\kappa)$ contingency tables with row sums $(m, n-m)$ and column sums κ, that is, to the pairs of weak compositions ρ of m and σ of $n-m$ with $\rho_i+\sigma_i = \kappa_i$. For the table (ρ,σ), the intersection of $S_{(m,n-m)}$ with the conjugate of S_κ by the distinguished double coset representative is $S_\rho \times S_\sigma$, and the index representation restricts to the index representation. So

$$\mathrm{Res}^{H_n}_{H_m\otimes H_{n-m}}\, \eta^\kappa_q = \sum_{\rho+\sigma=\kappa} \eta^\rho_q \otimes \eta^\sigma_q,$$

where η^ρ_q for a weak composition means $\eta^{\mathrm{sort}(\rho)}_q$. This is the same expansion as $\Delta h_\kappa = \sum_{\rho+\sigma=\kappa} h_\rho \otimes h_\sigma$. Now ϕ^λ_q is the same integer combination of the η^κ_q as m_λ is of the h_κ, so $\mathrm{Res}\,\phi^\lambda_q$ is the combination of the $\eta^\rho_q \otimes \eta^\sigma_q$ with the coefficients of Δm_λ in the basis $\{h_\rho \otimes h_\sigma\}$. Since $\Delta m_\lambda = \sum_{\mu\cup\nu=\lambda} m_\mu \otimes m_\nu$, this gives $\mathrm{Res}\,\phi^\lambda_q = \sum_{\mu\cup\nu=\lambda} \phi^\mu_q \otimes \phi^\nu_q$ and the first formula. For the second use $\phi^\nu_q(T_e) = \phi^\nu(e) = \delta_{\nu,1^k}$, as in Lemma 4.1; symmetry and unimodality are preserved because $\ell(w \times e_k) = \ell(w)$. □

So Theorem 1.3 follows from the case $n = 10$, which is Theorem 7.1 below.

Example 4.A. For $w = 231 \in S_3$, adding a fixed point gives $w \times e_1 = 2314 \in S_4$, and

$$c_{(3),231} = c_{(3,1),2314} = 3, \qquad c_{(2,1),231} = c_{(2,1,1),2314} = 1, \qquad c_{(1^3),231} = c_{(1^4),2314} = 0.$$

The remaining shape $(2,2) \vdash 4$ is not of the form $\lambda \cup 1$, and indeed $c_{(2,2),2314} = 0$, as Lemma 2.1 requires: no $\mu \vdash 3$ has $\mu \cup (1) = (2,2)$, so the sum is empty.

5. Extremal pairs

Lemma 5.1 ([12, (2.3.g)])**.** *If $s \in D_L(w)$ then $P_{x,w} = P_{sx,w}$ for all x. If $t \in D_R(w)$ then $P_{x,w} = P_{xt,w}$ for all x.*

Let $I = D_L(w)$ and $J = D_R(w)$, and let W_I, W_J be the corresponding Young subgroups. By Lemma 5.1, $P_{x,w}$ is constant on each double coset $W_I x W_J$. The lower interval $\{x \le w\}$ is a union of such double cosets, by the lifting property. So it suffices to know $P_{x,w}$ for x maximal in its double coset. These are exactly the $x \le w$ with $D_L(x) \supseteq I$ and $D_R(x) \supseteq J$, and we call (x, w) an *extremal pair*. Write $\hat{y} = \max(W_I y W_J)$, so that $P_{y,w} = P_{\hat{y},w}$.

Proposition 5.2. *S_9 has* 33,850,056 *extremal pairs and S_{10} has* 1,252,173,030*. There are* 169,438 *distinct pairs $(D_L(w), D_R(w))$ in S_{10}.*

Double cosets as contingency tables. Let α be the composition of positions cut by J and β the composition of values cut by I. Right multiplication by W_J permutes positions within the blocks of α, and left multiplication by W_I permutes values within the blocks of β. So the double coset of y is determined by its table $T(y)_{ab} = \#\{i \in \alpha_a : \ y(i) \in \beta_b\}$, a non-negative integer matrix with row sums α and column sums β, and every such matrix occurs. The double coset with table T has size $\prod_a \alpha_a! \prod_b \beta_b! / \prod_{a,b} T_{ab}!$. Its maximal element is obtained as follows: in each position block, the values appear in decreasing order; each position block takes, from each value block b, its T_{ab} largest unused values; and earlier position blocks receive larger values.

Lemma 5.3. *Number the position blocks $a = 1, \dots, p$ from the top and the value blocks $b = 1, \dots, r$ in increasing order of values, and put $i_a = \alpha_1 + \cdots + \alpha_a$ and $j_b = \beta_1 + \cdots + \beta_{b-1} + 1$, the last position of block a and the smallest value of block b. Let x and w be maximal in their (W_I, W_J)-double cosets, with tables $T(x)$ and $T(w)$. Then $x \le w$ iff the corner sums $S_T(a,b) = \sum_{a' \le a,\ b' \ge b} T_{a'b'}$ satisfy $S_{T(x)}(a,b) \le S_{T(w)}(a,b)$ for all $1 \le a < p$ and $1 < b \le r$. (For $a = p$ or $b = 1$ the corner sum is a sum of margins, the same for x and w.)*

Proof. By the tableau criterion, $x \le w$ iff $r_x(i,j) \le r_w(i,j)$ for all i, j, where $r_y(i,j) = \#\{i' \le i : y(i') \ge j\}$. At block corners, $r_y(i_a, j_b) = S_{T(y)}(a,b)$, which proves necessity.

For sufficiency, let $i_0 < i \le i_1$ lie in a position block. For a maximal element the values in the block decrease, so the positions of the block with value $\ge j$ come first. Hence $r_y(i,j) = \min\big(r_y(i_0,j) + (i - i_0),\ r_y(i_1,j)\big)$. Similarly, let j lie in a value block with values $j_0 < \cdots < j_1$, and let i_0 be a position-block boundary. The values of that block used by positions $\le i_0$ are its largest ones, so $r_y(i_0,j) = \min\big(r_y(i_0, j_1+1) + (j_1 + 1 - j),\ r_y(i_0, j_0)\big)$, where $r_y(i_0, j_1+1)$ and $r_y(i_0, j_0)$ are values at block corners. These expressions are monotone in the corner values and identical for x and w, so the corner inequalities propagate to all (i,j) (Figure 1). □

So the extremal $x \le w$ are enumerated directly, as the tables whose corner sums are dominated by those of $T(w)$, without listing the Bruhat interval.

Example 5.A. Let $w = 3412$. Then $D_L(w) = D_R(w) = \{s_2\}$, so $\alpha = \beta = (1,2,1)$: the position blocks are $\{1\}, \{2,3\}, \{4\}$ and so are the value blocks. The double cosets of S_4 correspond to the 3×3 tables with row and column sums $(1,2,1)$, and there are seven of them. Their sizes are $4/\prod T_{ab}!$, which sum to 24. The table lists each maximal element, its table, the size of its double coset, and the corner sums $S_T(a,b)$ for $a \in \{1,2\}$, $b \in \{2,3\}$; the other corner sums are fixed by the margins.

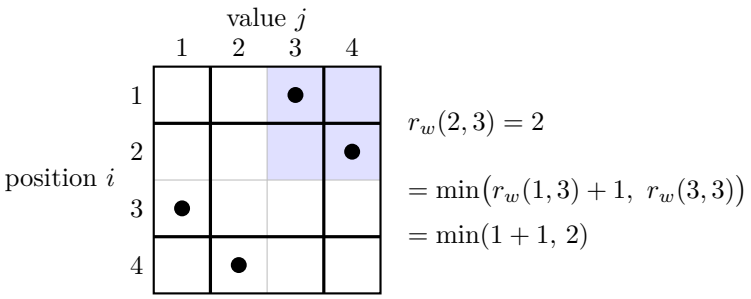


FIGURE 1. The permutation $w = 3412$ (a dot in row i, column $w(i)$) with the block grid of Example 5.A drawn thick. The shaded region counts $r_w(2,3) = \#\{i' \le 2 : w(i') \ge 3\}$. Row 2 is inside the position block $\{2,3\}$, and $r_w(2,3)$ is determined by the corner values $r_w(1,3)$ and $r_w(3,3)$, as in the proof of Lemma 5.3.

maximal element	T	size	$\begin{pmatrix} S_T(1,2) & S_T(1,3) \\ S_T(2,2) & S_T(2,3) \end{pmatrix}$	$\le w$
1324	$\left(\begin{smallmatrix}1&0&0\\0&2&0\\0&0&1\end{smallmatrix}\right)$	2	$\left(\begin{smallmatrix}0&0\\2&0\end{smallmatrix}\right)$	yes
1432	$\left(\begin{smallmatrix}1&0&0\\0&1&1\\0&1&0\end{smallmatrix}\right)$	4	$\left(\begin{smallmatrix}0&0\\2&1\end{smallmatrix}\right)$	yes
3214	$\left(\begin{smallmatrix}0&1&0\\1&1&0\\0&0&1\end{smallmatrix}\right)$	4	$\left(\begin{smallmatrix}1&0\\2&0\end{smallmatrix}\right)$	yes
3412	$\left(\begin{smallmatrix}0&1&0\\1&0&1\\0&1&0\end{smallmatrix}\right)$	4	$\left(\begin{smallmatrix}1&0\\2&1\end{smallmatrix}\right)$	$= w$
3421	$\left(\begin{smallmatrix}0&1&0\\0&1&1\\1&0&0\end{smallmatrix}\right)$	4	$\left(\begin{smallmatrix}1&0\\\mathbf{3}&1\end{smallmatrix}\right)$	no
4312	$\left(\begin{smallmatrix}0&0&1\\1&1&0\\0&1&0\end{smallmatrix}\right)$	4	$\left(\begin{smallmatrix}1&\mathbf{1}\\2&1\end{smallmatrix}\right)$	no
4321	$\left(\begin{smallmatrix}0&0&1\\0&2&0\\1&0&0\end{smallmatrix}\right)$	2	$\left(\begin{smallmatrix}1&\mathbf{1}\\\mathbf{3}&1\end{smallmatrix}\right)$	no

By Lemma 5.3, a maximal element lies below w iff its corner sums are at most those of w; the entries in bold are the ones that exceed them. So w has four extremal pairs, with $x = 1324, 1432, 3214, 3412$, while its Bruhat interval has 14 elements.

6. The recursion and the coefficients

Pick $s \in D_L(w)$ and put $v = sw$. For extremal $x \le w$ we have $sx < x$, and the standard recursion [12] reads

$$P_{x,w} = P_{sx,v} + q\,P_{x,v} - \sum_{z \in M} \mu(z,v)\, q^{(\ell(w)-\ell(z))/2} P_{x,z}, \qquad M = \{z < v :\ sz < z,\ \mu(z,v) \ne 0\}. \tag{2}$$

Each $P_{y,u}$ on the right is looked up as $P_{\hat{y},u}$ (with hats for the descents of u) in the sorted list of extremal pairs of u. If $\hat{y}$ is not in the list then $y \not\le u$ and $P_{y,u} = 0$. By the descent theorem [12, (2.3.e)], if $\mu(z,v) \ne 0$ and $\ell(v) - \ell(z) > 1$ then $D_L(v) \subseteq D_L(z)$ and $D_R(v) \subseteq D_R(z)$. So every $z \in M$ is either a Bruhat cover of v, with $\mu = 1$, or extremal for v, with $\mu(z,v)$ read off the stored polynomial. Elements are processed by increasing length, and equal polynomials share an identifier in a hash table.

Example 6.A. Continue Example 5.A with $w = 3412$ and $s = s_2$, so $v = s_2 w = 2413$. Here $D_L(v) = \{s_1, s_3\}$ and $D_R(v) = \{s_2\}$, and the only extremal pair of v is (v,v); there is no $z < v$

with $s_2 z < z$ and $\mu(z,v) \neq 0$, so $M = \emptyset$ and (2) reads $P_{x,w} = P_{sx,v} + q\,P_{x,v}$. For the four extremal x:

- $x = 1324$: $sx = 1234$, and both $\widehat{1234}$ and $\widehat{1324}$ (for the descents of v) equal $2413 = v$. So $P_{x,w} = 1 + q$.
- $x = 1432$: $sx = 1423$ with $\widehat{1423} = v$, so $P_{sx,v} = 1$. But $\widehat{1432} = 2431$ is longer than v, so it is not in the list of v, and $P_{x,v} = 0$. So $P_{x,w} = 1$.
- $x = 3214$ and $x = 3412$ behave in the same way, with $\hat{x} = 4213$ and 4321 not below v, and $P_{x,w} = 1$.

6.1. **Coefficients.** By (1) and Lemma 5.1,

(3)
$$c_{\lambda,w} = \sum_{v \le w} P_{v,w}(1)\,\phi^\lambda(v) = \sum_{x \text{ extremal for } w} P_{x,w}(1)\,G_\lambda(I,J,x), \qquad G_\lambda(I,J,x) = \sum_{v \in W_I x W_J} \phi^\lambda(v).$$

G_λ depends on w only through (I,J). For each of the 169,438 classes we run once through S_{10} by depth-first search. The double coset of each v is identified by an additive hash of its table, updated one position at a time. The number of v assigned to each double coset is then checked against the size formula of Section 5, which rules out hash collisions. For $n \le 10$ all values $\phi^\lambda(v)$ are integers, and $|c_{\lambda,w}| \le n!\,\max|\phi^\lambda|\,\max_{x,w} P_{x,w}(1) < 2^{62}$, so all arithmetic is exact in 64-bit integers.

Example 6.B. For $w = 3412$, the four double cosets of Example 5.A that lie below w give the following values of $G_\lambda(I,J,x)$, for $\lambda = (4),(3,1),(2,2),(2,1,1),(1^4)$:

$$G(1324) = (0,0,0,2,0), \quad G(1432) = G(3214) = (0,6,0,-2,0), \quad G(3412) = (8,-8,4,0,0).$$

With the weights $P_{x,w}(1) = 2,1,1,1$ from Example 6.A, (3) gives

$$c_{\cdot,3412} = 2\,(0,0,0,2,0) + 2\,(0,6,0,-2,0) + (8,-8,4,0,0) = (8,4,4,0,0).$$

The individual G_λ can be negative, and for $\lambda = (2,1,1)$ the coefficient vanishes only because of the weight $P_{1324,3412}(1) = 2$.

6.2. **Graded coefficients.** At generic q the same double-coset reduction gives

$$c_{\lambda,w}(q) = \sum_{x \text{ extremal for } w} P_{x,w}(q)\,G_\lambda(I,J,x)(q), \qquad G_\lambda(I,J,x)(q) = \sum_{v \in W_I x W_J} \phi^\lambda_q(T_v), \tag{4}$$

with the same depth-first runs and table hashes as at $q = 1$; only the accumulated quantity is now a vector of polynomials. The traces $\phi^\lambda_q(T_v)$ come from the class polynomials of Geck and Pfeiffer [8, Ch. 8]. For every trace τ of H_n: if $\ell(sus) = \ell(u)$ then $\tau(T_u) = \tau(T_{sus})$; if $\ell(sus) = \ell(u) - 2$ then $\tau(T_u) = (q-1)\tau(T_{su}) + q\,\tau(T_{sus})$; and every u reaches a minimal-length element of its class by cyclic shifts that do not increase length, all minimal-length elements of a class having the same trace [8, Thms. 3.2.9, 8.2.3]. So $T_v \equiv \sum_\rho f_{v,\rho}(q)\,T_{c_\rho}$ modulo $[H_n,H_n]$ with integer polynomials $f_{v,\rho}$, computed by increasing length. For $h \in H_n$ write $\mathrm{ch}(h) = \sum_a \chi^a_q(h)\,s_a$, where χ^a_q is the irreducible character of H_n that specialises to χ^a at $q = 1$; the index and sign representations are $T_s \mapsto q$ and $T_s \mapsto -1$, with characters $\chi^{(n)}_q$ and $\chi^{(1^n)}_q$. Since $\chi^a_q = \sum_\lambda K_{a,\lambda}\phi^\lambda_q$ and $h_\lambda = \sum_a K_{a,\lambda} s_a$, we get $\mathrm{ch}(h) = \sum_\lambda \phi^\lambda_q(h)\,h_\lambda$: the coefficient of h_λ in $\mathrm{ch}(h)$ is $\phi^\lambda_q(h)$. At the minimal-length element c_ρ, a product of Coxeter elements of consecutive blocks of sizes $\rho_1,\rho_2,\dots$, one has $\mathrm{ch}(T_{c_\rho}) = \prod_k \sum_j (-1)^j q^{\rho_k - 1 - j} s_{(\rho_k - j, 1^j)}$ [17], which at $q = 1$ is the power sum p_ρ. Expanding in the h basis gives $\phi^\lambda_q(T_{c_\rho})$, exactly, in rational arithmetic. Since $\deg \phi^\lambda_q(T_v) \le \ell(v)$ and $\deg P_{v,w} \le (\ell(w) - \ell(v) - 1)/2$ for $v < w$, while $P_{w,w} = 1$, every $c_{\lambda,w}(q)$ has degree at most $\ell(w)$. For $n = 10$ the traces and the output each take about 56 GB, and the shapes are processed in batches. On 192 cores the Kazhdan–Lusztig table takes about 100 s in both the $q = 1$ and

the graded run, while the graded coefficient step takes about 38 times as long as at $q = 1$ (84 minutes against 131 s), since each accumulated G_λ is a polynomial of degree at most 45 instead of an integer.

Example 6.C. In S_3 the classes are represented by $c_{(1^3)} = e$, $c_{(2,1)} = s_1$ and $c_{(3)} = s_1 s_2$, and

$$\mathrm{ch}(T_e) = h_1^3, \qquad \mathrm{ch}(T_{s_1}) = (qs_2 - s_{11})s_1 = (1+q)h_{21} - h_{111},$$
$$\mathrm{ch}(T_{s_1 s_2}) = q^2 s_3 - qs_{21} + s_{111} = (1+q+q^2)h_3 - (2+q)h_{21} + h_{111}.$$

The other elements are reduced to these. The elements $213 = s_1$ and $132 = s_2$ both have minimal length in their class, and 312 is a cyclic shift of $231 = s_1 s_2$ of the same length, so they share the traces above. For $w_0 = 321$, conjugating by $s = s_1$ gives $sw_0 s = s_2$, of length $\ell(w_0) - 2$, so $\tau(T_{321}) = (q-1)\tau(T_{312}) + q\,\tau(T_{132})$ for every trace τ. The values $\phi_q^\lambda(T_v)$ are:

v	123	213, 132	231, 312	321
$\phi_q^{(3)}(T_v)$	0	0	$1+q+q^2$	q^3-1
$\phi_q^{(2,1)}(T_v)$	0	$1+q$	$-2-q$	2
$\phi_q^{(1^3)}(T_v)$	1	-1	1	-1

At $q = 1$ the rows are the class functions ϕ^λ. The lower interval of $w = 231$ is $\{123, 213, 132, 231\}$, with all $P_{v,w} = 1$, so

$$c_{\cdot,231}(q) = \big(1+q+q^2,\ q,\ 0\big) \qquad (\lambda = (3), (2,1), (1^3)),$$

which specialises to the values $3, 1, 0$ of Example 4.A. Summing over all of S_3 gives $c_{\cdot,321}(q) = \big((1+q)(1+q+q^2), 0, 0\big) = \big([3]_q!, 0, 0\big)$, the graded form of Lemma 3.1(3). The same table comes out of the induced modules of check 12 below, which use neither class polynomials nor [17].

7. Verification and results

Theorem 7.1. *For every $\lambda \vdash 10$ and every $w \in S_{10}$, $c_{\lambda,w}(q) \in \mathbb{Z}_{\ge 0}[q]$, and it is symmetric and unimodal about $q^{\ell(w)/2}$.*

The computation of Sections 5 and 6 gives every $c_{\lambda,w}(q)$. None has a negative coefficient, none is asymmetric and none fails to be unimodal. S_{10} has 981,174 distinct Kazhdan–Lusztig polynomials, of degree at most 12; the largest coefficient of any $c_{\lambda,w}(q)$ is 35,419,702. Table 2 gives the support and range of the non-zero values at $q = 1$. The ungraded computation took 279 seconds on a 192-core machine, with a peak memory of 15.5 GB; the graded one took 1 h 31 min, with a peak of 125 GB, almost all of it in accumulating (4). All arithmetic is exact in 64-bit integers, by the a priori bounds at the end of this section. We ran the following checks.

(1) *Sanity of the polynomials.* Every stored $P_{x,w}$ has constant term 1 and non-negative coefficients, $P_{w,w} = 1$, and $\deg P_{x,w} \le (\ell(w) - \ell(x) - 1)/2$ for $x < w$.

(2) *Independent software.* For every $n \le 10$, every extremal pair, including all 1,252,173,030 in S_{10}, was compared with du Cloux's Coxeter3 [4], with no discrepancies. This took 2.4 hours on 190 cores.

(3) *J-values.* For all $2^9 = 512$ compositions α of 10 and all λ, $c_{\lambda,w_0(\alpha)} = \alpha!\,\delta_{\lambda,\mathrm{sort}(\alpha)}$ (Lemma 3.1(3)).

(4) *Symmetries.* $c_{\lambda,w} = c_{\lambda,w^{-1}} = c_{\lambda,w_0 w w_0}$ for all λ, w. The left-descent recursion does not build these in.

(5) *Kazhdan–Lusztig inversion.* For 2000 random $u \in S_{10}$ and every λ, the coefficient of $x_{1,u(1)} \cdots x_{n,u(n)}$ in $\sum_w c_{\lambda,w}\,\mathrm{Imm}_w$ equals $\phi^\lambda(u)$. This tests the coefficients against the inversion formula, using the stored polynomials at all pairs $x \le w$, not only extremal ones. The check is exact over $\mathbb{Z}$: the sum $\sum |P(1)|\,|c_{\lambda,w}|$ bounds every partial sum and is

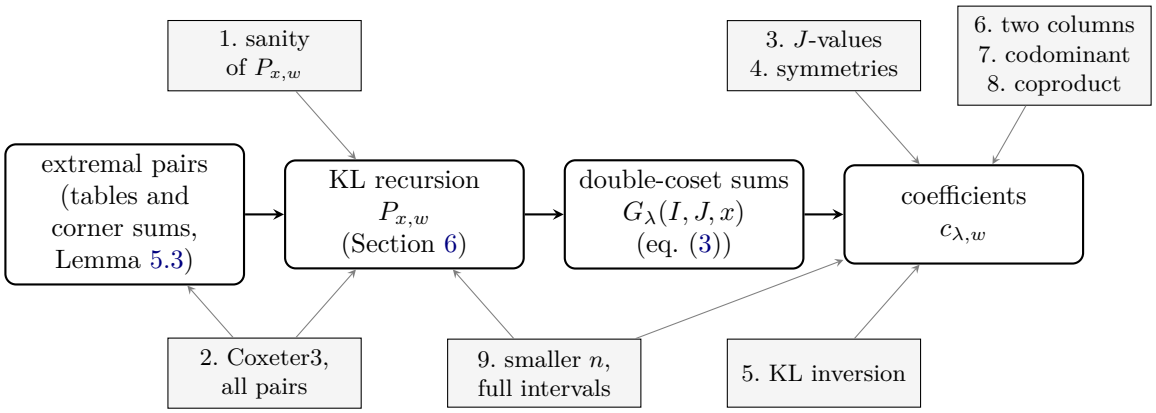


FIGURE 2. The computation of Theorem 7.1 (top row) and the stages tested by the checks of Section 7. Check 2 compares every extremal pair, so it also tests that the enumerated pairs are the right ones. Checks 3, 4 and 6–8 test the output against facts proved independently of the computation.

below 2^{41}, so no 64-bit partial sum wraps around, and the sums were also accumulated modulo a prime $p < 2^{31}$.

(6) *Two columns.* For $\lambda_1 \le 2$, $c_{\lambda,u} = 0$ whenever u contains 321, a theorem that follows from Fan and Green's description of the Temperley–Lieb quotient [6] (see [13, Lemma 8.1]).

(7) *Codominant elements.* For each of the 16,796 codominant (312-avoiding) w, $c_{\lambda,w} = \mathrm{Imm}_\lambda(A_w)$, where A_w is the 0/1 matrix with $(A_w)_{ij} = [j \le \max(w(1),\dots,w(i))]$ [13, Prop. 7.1]. The right-hand side was evaluated without any Kazhdan–Lusztig computation. Check 13 covers the same w for every q, against reference values that do not depend on [13].

(8) *Coproduct rule.* For all 799,475 decomposable $w = w_1 \times w_2$ and all λ (33,577,950 coefficients), $c_{\lambda,w}$ agrees with Lemma 2.1 applied to independent runs for S_m, $m < 10$.

(9) *Smaller n.* For $n \le 9$ the engine reproduces every $c_{\lambda,w}$ of an independent full-interval engine, which stores all 10,501,351,657 Bruhat pairs of S_9. For small n every stored $P_{x,w}$ agrees with a reference implementation and with SageMath.

(10) *Graded output at $q = 1$.* $c_{\lambda,w}(1)$ equals the value of the ungraded run for every λ, w.

(11) *Graded symmetries and stability.* $c_{\lambda,w}(q) = c_{\lambda,w^{-1}}(q) = c_{\lambda,w_0ww_0}(q)$ for all λ, w, and Lemma 4.3 holds exactly against a separate run for S_9. The method builds in neither.

(12) *Traces from the Hecke relations.*

(a) *Induced modules, $n \le 6$.* For $n \le 6$ and $q = 2, 3$, every $c_{\lambda,w}(q)$ equals the trace of $\sum_v P_{v,w}(q)T_v$ on the induced modules $M^\mu = H_n \otimes_{H_\mu}$ (index), built from the defining relations alone, without class polynomials or character formulas, and with Kazhdan–Lusztig polynomials from a separate implementation of the standard recursion. For $n \le 5$ the comparison is exact over $\mathbb{Z}$; for $n = 6$ (modules of dimension up to 720) it is made modulo the primes $2^{20} - 3$ and $2^{20} - 5$, with no mismatch.

(b) *Induced modules, $n = 10$.* The six modules M^μ with $\mu = (9,1), (8,2), (8,1,1), (7,3), (6,4), (5,5)$, of dimension at most 252, are traversed depth first along a spanning tree of the weak order. For every $v \in S_{10}$ the trace $\eta^\mu_q(T_v)$ agrees with $\sum_\lambda N_{\mu\lambda}\phi^\lambda_q(T_v)$ computed from the class polynomials, where $h_\mu = \sum_\lambda N_{\mu\lambda}m_\lambda$.

(c) *Seminormal modules, $n = 10$.* Every trace column is tested in the same way through the 42 irreducible characters. Hoefsmit's seminormal form [11, 18] gives explicit matrices for the T_s on the irreducible modules S^μ, of dimension at most 768, from the combinatorics of standard tableaux; they satisfy the quadratic and braid relations

(checked for every shape of size at most 8). Traversed in the same way, $\chi_q^\mu(T_v)$ agrees with $\sum_\lambda K_{\mu\lambda}\phi_q^\lambda(T_v)$ for every $\mu \vdash 10$ and every $v \in S_{10}$, where $s_\mu = \sum_\lambda K_{\mu\lambda} m_\lambda$. Since the Kostka matrix is unitriangular, this tests each of the 42 traces $\phi_q^\lambda(T_v)$ at every v.

(d) *Sampled w, $n = 10$.* For 300 random $w \in S_{10}$ containing 312, with $P_{v,w} = P_{v^*,w}$ from the extremal pairs listed by Coxeter3 (v^* the maximum of $W_{D_L(w)} v W_{D_R(w)}$, Lemma 5.1), $\sum_\lambda N_{\mu\lambda} c_{\lambda,w}(q)$ agrees with $\sum_v P_{v,w}(q)\eta_q^\mu(T_v)$ for the six μ, and the stored $c_{\lambda,w}(q)$ agree with $\sum_v P_{v,w}(q)\phi_q^\lambda(T_v)$ for all 42 shapes. The last comparison takes the traces $\phi_q^\lambda(T_v)$ as given and tests only the step (4); the others use no class polynomial. The comparisons at $n = 10$ are made at $q = 2, 3$ modulo the prime $2^{31} - 19$, with no mismatch.

(13) *Chromatic quasisymmetric functions.* For each of the 16,796 codominant w and all 42 shapes, $c_{\lambda,w}(q)$ equals the coefficient of e_λ in the chromatic quasisymmetric function of the indifference graph of w, computed from proper colourings without any Kazhdan–Lusztig or Hecke algebra input.

Check 2 covers the polynomials, and checks 3–8 the coefficient step (3), partly with no Kazhdan–Lusztig input at all (checks 3, 6 and 7). Checks 10–13 cover the graded step (4); check 13 is independent of the whole pipeline. Figure 2 shows which stage each check tests, and Table 1 which code each check shares with the computation.

Independence. At $n = 10$ the stages E, K, S and T (at $q = 1$) are each tested by a check that shares none of their code: E and K by check 2, S by checks 5 and 7, and T at $q = 1$ by check 10. The table Φ is tested against a theorem by check 3, and recomputed from scratch in check 12 for $n \leq 6$. Agreement between programs that share code could hide a common error, and the table shows where this matters. At $q = 2$ and $q = 3$, modulo $2^{31} - 19$, T is tested independently at every $v \in S_{10}$ by check 12, through all 42 irreducible characters; it is also tested by check 11. These are evaluations at two values of q, not identities of polynomials in q. The weakest point is S at generic q for non-codominant w, the assembly (4) of the polynomials $c_{\lambda,w}(q)$ from the traces. At $q = 1$ the same step is tested independently by check 5, at 2000 random u, and by check 7, at all 16,796 codominant w. At generic q it is tested by check 11 and by the symmetry and unimodality counts, which the method does not build in, and independently only end to end, at 300 sampled w and at $q = 2, 3$ (check 12). An exhaustive independent test would need the sum over all $\sim 10^{12}$ Bruhat pairs of S_{10}. The checks that fix the conventions for T_s, C'_w and the Frobenius map independently are check 12, from the defining relations ($n \leq 6$ for every w, $n = 10$ as above), and check 13, from colourings (codominant w, $n \leq 10$); a convention error would have to pass both.

Integer safety. No intermediate value can leave the range of signed 64-bit integers, by three bounds that are a priori except for two numbers read off the finished tables, as explained below. (i) Let $A(u)$ be the sum of the absolute values of the coefficients of all the class polynomials $f_{u,\rho}$. No element of length at most 2 admits a conjugation $u \mapsto sus$ with $\ell(sus) = \ell(u) - 2$, so these elements have $A(u) = 1$; cyclic shifts do not change A, and the second rule of Section 6.2 gives $A(u) \leq 2A(su) + A(sus)$. Hence $A(u) \leq M_{\ell(u)}$ with $M_0 = M_1 = M_2 = 1$ and $M_\ell = 2M_{\ell-1} + M_{\ell-2}$, and the same bounds every partial sum of the recursion. For $n = 10$, $M_{45} \approx 3.48 \times 10^{16} < 2^{61}$. (ii) Every partial sum of $\phi_q^\lambda(T_v) = \sum_\rho f_{v,\rho}\, \phi_q^\lambda(T_{c_\rho})$ is at most $M_{45}\beta$, where $\beta = 120$ is the largest sum of the absolute values of the coefficients of a $\phi_q^\lambda(T_{c_\rho})$, computed in rational arithmetic. So it is at most $4.17 \times 10^{18} < 2^{63}$. (iii) In (4) the double-coset sums are at most $n!\, m$, and every partial sum is at most $m \cdot n! \cdot \max_{x,w} P_{x,w}(1) = 6.68 \times 10^{13}$, where m is the largest sum of the absolute values of the coefficients of a trace $\phi_q^\lambda(T_v)$. The numbers m and $\max P_{x,w}(1)$ are read off the finished tables, which are exact by (i), (ii) and check 2. Degrees are guarded separately:

check	code shared	compared with	stages tested
1	E, K	constant term and degree bound of KL polynomials	K
2	none	Coxeter3 (independent C++), every extremal pair and its x-list	E, K
3	E, K, Φ, S	Lemma 3.1(3), a theorem	Φ, S
4	E, K, Φ, S	symmetries of $P_{v,w}$, not built into the recursion	K, S
5	K, Φ	identity (1) at all pairs $x \le w$, without the double-coset sums	S
6	E, K, Φ, S	Fan–Green [6], a theorem	K, S
7	Φ	$\mathrm{Imm}_\lambda(A_w)$, no Kazhdan–Lusztig input	E, K, S (codominant w)
8	E, K, Φ, S	Lemma 2.1, separate runs for S_m, $m < 10$	consistency in n
9	Φ	full-interval engine (separate code); for small n a reference implementation and SageMath	E, K, S ($n \le 9$)
10	E, K, S	the ungraded run, which uses Φ instead of T	T at $q = 1$
11	E, K, S, T	symmetries and Lemma 4.3, not built in	T
12	none; at $n = 10$ Lemma 5.1, which S also uses, and T as input to one comparison	traces on induced modules built from the Hecke relations, and at $n = 10$ on Hoefsmit's seminormal modules; KL polynomials by the standard recursion ($n \le 6$) or Coxeter3 ($n = 10$)	all ($n \le 6$); at $n = 10$, T at every v (all 42 characters) and all stages at 300 sampled w
13	none	chromatic quasisymmetric functions from colourings, via the dictionary of [2]	all (codominant w)

TABLE 1. Code shared by each check of Section 7 with the computation. The stages are: E, the enumeration of extremal pairs (Lemma 5.3); K, the Kazhdan–Lusztig recursion (Section 6); Φ, the table of ϕ^λ on conjugacy classes; S, the double-coset sums (3) and (4); T, the traces $\phi^\lambda_q(T_v)$ from class polynomials (Section 6.2).

a coefficient beyond the stored degree stops the run. The bound at $q = 1$ in Section 6.1 is the analogue of (iii). For comparison, the largest coefficient that occurs is 35,419,702.

8. Consequences for $n \le 10$

Proof of Theorem 1.3 and Corollary 1.4. Theorem 7.1 and Corollary 4.2 give Theorem 1.3. Then $C_{\mathrm{KL}} = C_J$ by Theorem 3.2, and the chain $C_{\mathrm{KL}} \subseteq C_{\mathrm{TNN}} \subseteq C_J$ gives $C_{\mathrm{KL}} = C_{\mathrm{TNN}} = C_J$. □

8.1. **At generic q.** Positivity at q is coefficientwise, so the analogues of the cones are $\mathbb{N}[q]$-semimodules of traces of H_n with values in $\mathbb{Z}[q]$. Let $C_{\mathrm{KL}}(q)$ be the set of traces τ with $\tau(q^{\ell(w)/2}C'_w) \in \mathbb{N}[q]$ for all $w \in S_n$, and let $C_J(q) = \sum_\lambda \mathbb{N}[q]\,\phi^\lambda_q$. Since $q^{\ell(w_0(\mu))/2}C'_{w_0(\mu)} = \sum_{v \in S_\mu} T_v$ and ϕ^λ_q takes the value $\delta_{\lambda\mu}[\mu]_q!$ there ($[\mu]_q! = \prod_i [\mu_i]_q!$; Haiman [9, Prop. 4.1(1)]), the coordinates of $\tau = \sum_\lambda a_\lambda \phi^\lambda_q$ are $a_\mu = \tau\big(\sum_{v \in S_\mu} T_v\big)/[\mu]_q!$, and $\tau \in C_J(q)$ iff all of them lie in $\mathbb{N}[q]$. The positivity part of Conjecture 1.1 is the inclusion $C_J(q) \subseteq C_{\mathrm{KL}}(q)$, and Theorem 1.3 gives it for $n \le 10$.

λ	$\#\{w : c_{\lambda,w} > 0\}$	range	λ	$\#\{w : c_{\lambda,w} > 0\}$	range
10	2,829,325	10–384,025,920	$4\,3^2$	1,446,300	3–192,364
9 1	3,376,010	8–253,386,624	4 3 2 1	1,844,393	1–209,952
8 2	2,887,431	16–109,171,840	$4\,3\,1^3$	659,730	2–17,984
$8\,1^2$	3,431,287	7–37,088,768	$4\,2^3$	397,450	4–17,248
7 3	2,847,535	21–51,336,714	$4\,2^2\,1^2$	527,279	2–11,572
7 2 1	3,441,650	7–25,228,512	$4\,2\,1^4$	103,683	2–736
$7\,1^3$	2,511,423	6–1,264,516	$4\,1^6$	5,358	3–160
6 4	2,803,637	24–24,591,232	$3^3\,1$	500,625	2–17,020
6 3 1	3,362,247	12–11,369,056	$3^2\,2^2$	308,894	2–11,588
$6\,2^2$	2,458,248	8–2,578,752	$3^2\,2\,1^2$	367,081	1–4,836
$6\,2\,1^2$	2,709,706	4–867,808	$3^2\,1^4$	49,280	2–440
$6\,1^4$	680,307	5–48,560	$3\,2^3\,1$	133,373	1–1,340
5^2	2,534,300	15–9,236,960	$3\,2^2\,1^3$	55,642	1–232
5 4 1	3,064,042	6–6,169,664	$3\,2\,1^5$	7,421	1–48
5 3 2	2,486,584	4–1,581,664	$3\,1^7$	309	2–8
$5\,3\,1^2$	2,542,958	4–464,880	2^5	5,662	2–32
$5\,2^2\,1$	1,954,375	3–240,900	$2^4\,1^2$	8,640	1–16
$5\,2\,1^3$	839,778	3–34,700	$2^3\,1^4$	2,167	1–8
$5\,1^5$	76,455	4–1,200	$2^2\,1^6$	301	1–4
$4^2\,2$	1,660,725	4–482,904	$2\,1^8$	25	1–2
$4^2\,1^2$	1,807,360	3–195,552	1^{10}	1	1–1

TABLE 2. Support size and range of the non-zero coefficients $c_{\lambda,w}$, $n = 10$.

Unlike at $q = 1$, this inclusion need not be an equality, and the evaluations at the $w_0(\mu)$ do not detect membership in $C_{\mathrm{KL}}(q)$. Let $C_J^{\mathrm{ev}}(q)$ be the set of traces with $\tau(\sum_{v\in S_\mu} T_v) \in \mathbb{N}[q]$ for all μ. Then $C_{\mathrm{KL}}(q) \subseteq C_J^{\mathrm{ev}}(q)$ for every n, and for $n \leq 10$

$$C_J(q) \subseteq C_{\mathrm{KL}}(q) \subseteq C_J^{\mathrm{ev}}(q).$$

The first inclusion is strict for $n = 2$: the trace $(1 - q + q^2)\phi_q^{(2)}$ takes the values 0 at T_e and $(1-q+q^2)(1+q) = 1+q^3$ at $q^{1/2}C'_s = T_e + T_s$, so it lies in $C_{\mathrm{KL}}(q)$, but its coordinate $1-q+q^2$ is not in $\mathbb{N}[q]$. The second is strict for every $n \geq 3$: the trace $\tau = (1-q+q^2)\phi_q^{(2,1^{n-2})}$ has a single non-zero evaluation at the $w_0(\mu)$, namely $(1-q+q^2)[2]_q = 1+q^3$, so $\tau \in C_J^{\mathrm{ev}}(q)$; but by Lemma 4.3 and Example 6.C, $c_{(2,1^{n-2}),\,231\times e_{n-3}}(q) = c_{(2,1),231}(q) = q$, so τ takes the value $q - q^2 + q^3$ there and $\tau \notin C_{\mathrm{KL}}(q)$. At $q = 1$ the difference disappears, because a real number a_μ with $\mu!\, a_\mu \geq 0$ is itself non-negative. So the graded form of Corollary 1.4 is the inclusion $C_J(q) \subseteq C_{\mathrm{KL}}(q)$ for $n \leq 10$, and nothing more. This inclusion still lifts every relation of the normalised order to q:

Corollary 8.1. *Let $n \leq 10$ and let $a, b \vdash n$ satisfy the criterion of Theorem 1.5. Then $\chi_q^a(q^{\ell(w)/2}C'_w)/f^a - \chi_q^b(q^{\ell(w)/2}C'_w)/f^b$ has non-negative rational coefficients for every $w \in S_n$.*

Proof. $\chi_q^a/f^a - \chi_q^b/f^b = \sum_\mu (K_{a,\mu}/f^a - K_{b,\mu}/f^b)\phi_q^\mu$, with non-negative coefficients by the criterion; evaluate at $q^{\ell(w)/2}C'_w$ and apply Theorem 1.3. □

8.2. **At $q = 1$.** In terms of irreducible characters, for $\phi = \sum_a \gamma_a \chi^a$ we have $d_\phi(J_\mu) = \mu! \sum_a \gamma_a K_{a,\mu}$. So for $n \leq 10$, every linear inequality between irreducible-character immanants on TNN_n is decided by a finite check on Kostka numbers. The same holds for inequalities known on positive semidefinite matrices, since each J_μ is positive semidefinite; for normalised characters this observation is Stembridge's [25, §3] (see [13, Prop. 6.5]).

8.3. **The order of normalised immanants.** For $a, b \vdash n$ write $a \succeq b$ if $N_a \geq N_b$ on TNN_n.

Proof of Theorem 1.5. $f^a f^b(N_a - N_b) = d_\phi$ with $\phi = f^b\chi^a - f^a\chi^b$, and $d_\phi(J_\mu) = \mu!\,(f^b K_{a,\mu} - f^a K_{b,\mu})$ by Lemma 3.1. So the criterion is $\phi \in C_J$, and $a \succeq b$ is $\phi \in C_{\mathrm{TNN}}$. They agree by Corollary 1.4. □

The necessity of the criterion holds for every n, since $J_\mu \in \mathrm{TNN}_n$; it is Stembridge's [25, Prop. 3.1], who notes that his conjecture gives the converse (see also [13, Prop. 6.3]). The content of Theorem 1.5 is the converse for $n \leq 10$.

Taking $\mu = b$ shows that $a \succeq b$ implies $K_{a,b} > 0$, so $\succeq$ is contained in dominance order for every n. Figure 3 shows the Hasse diagrams for $n \leq 7$, and Appendix A those for $n = 8, 9, 10$.

Example 8.A. Let $n = 4$, $a = (3,1)$ and $b = (2,2)$, so a dominates b, and $f^a = 3$, $f^b = 2$. For $\mu = (4), (3,1), (2,2), (2,1,1), (1^4)$,

$$\frac{K_{a,\mu}}{f^a} = \Big(0, \tfrac{1}{3}, \tfrac{1}{3}, \tfrac{2}{3}, 1\Big), \qquad \frac{K_{b,\mu}}{f^b} = \Big(0, 0, \tfrac{1}{2}, \tfrac{1}{2}, 1\Big).$$

The criterion for $a \succeq b$ fails at $\mu = (2,2)$, and indeed $N_a(J_{22}) = 4 \cdot \frac{1}{3} < N_b(J_{22}) = 4 \cdot \frac{1}{2}$. The criterion for $b \succeq a$ fails at $\mu = (3,1)$, where $N_b(J_{31}) = 0 < N_a(J_{31}) = 2$. So $(3,1)$ and $(2,2)$ are incomparable in $\succeq$, which is the first failure of dominance in Theorem 8.2(1).

Theorem 8.2. *Let $n \leq 10$.*

(1) *$\succeq$ equals dominance order iff $n \leq 3$. The first pair that is comparable in dominance but not in $\succeq$ is $(3,1)$ and $(2,2)$.*
(2) *$\succeq$ is a lattice iff $n \leq 5$. For $n = 6$, the elements $(4,2)$ and $(3,3)$ have two maximal common lower bounds, $(3,2,1)$ and $(2,2,2)$, and hence no meet (Figure 4).*
(3) *The hook chain $(n) \succ (n-1,1) \succ \cdots \succ (1^n)$ and the two-column chain $(2^k, 1^{n-2k}) \succ (2^{k-1}, 1^{n-2k+2})$ are chains of $\succeq$. The two-row chain $(n-k, k)$, $0 \leq k \leq n/2$, is a chain only for $n \in \{1,2,3,5,7\}$. For even $n \geq 4$ it fails at $(n/2+1, n/2-1) \not\succeq (n/2, n/2)$, and for $n = 9$ at $(6,3) \not\succeq (5,4)$.*
(4) *Conjugation $a \mapsto a'$ is not an anti-automorphism of $\succeq$ for $n \geq 4$.*

Proof. Finite computations with Theorem 1.5. □

Theorem 8.2 is a statement about $n \leq 10$ only, the range of the exhaustive computation; we know of no argument for larger n. A failure of a relation $a \succeq b$ is witnessed by some J_μ and so holds without Theorem 1.5, while every assertion that a relation holds, and hence the lattice property for $n \leq 5$, uses it.

For every n, (n) is the top and (1^n) the bottom element of $\succeq$, and the hook chain is a chain. On TNN_n the hook chain is due to Skandera [22, Thm. 27]; the companion paper gives a proof that both statements hold at the level of Kazhdan–Lusztig immanants [13, Prop. 6.4, Cor. 5.1]. Table 3 gives the size of the order.

n	3	4	5	6	7	8	9	10
partitions	3	5	7	11	15	22	30	42
comparable pairs in $\succeq$	3	9	20	48	96	198	375	723
comparable pairs in dominance	3	10	21	53	101	216	400	776
covers	2	5	7	16	21	40	55	102
height (longest chain)	3	4	6	7	10	12	15	16
width (largest antichain)	1	2	2	3	3	5	5	7

TABLE 3. Size of the normalised order $\succeq$ on partitions of n.

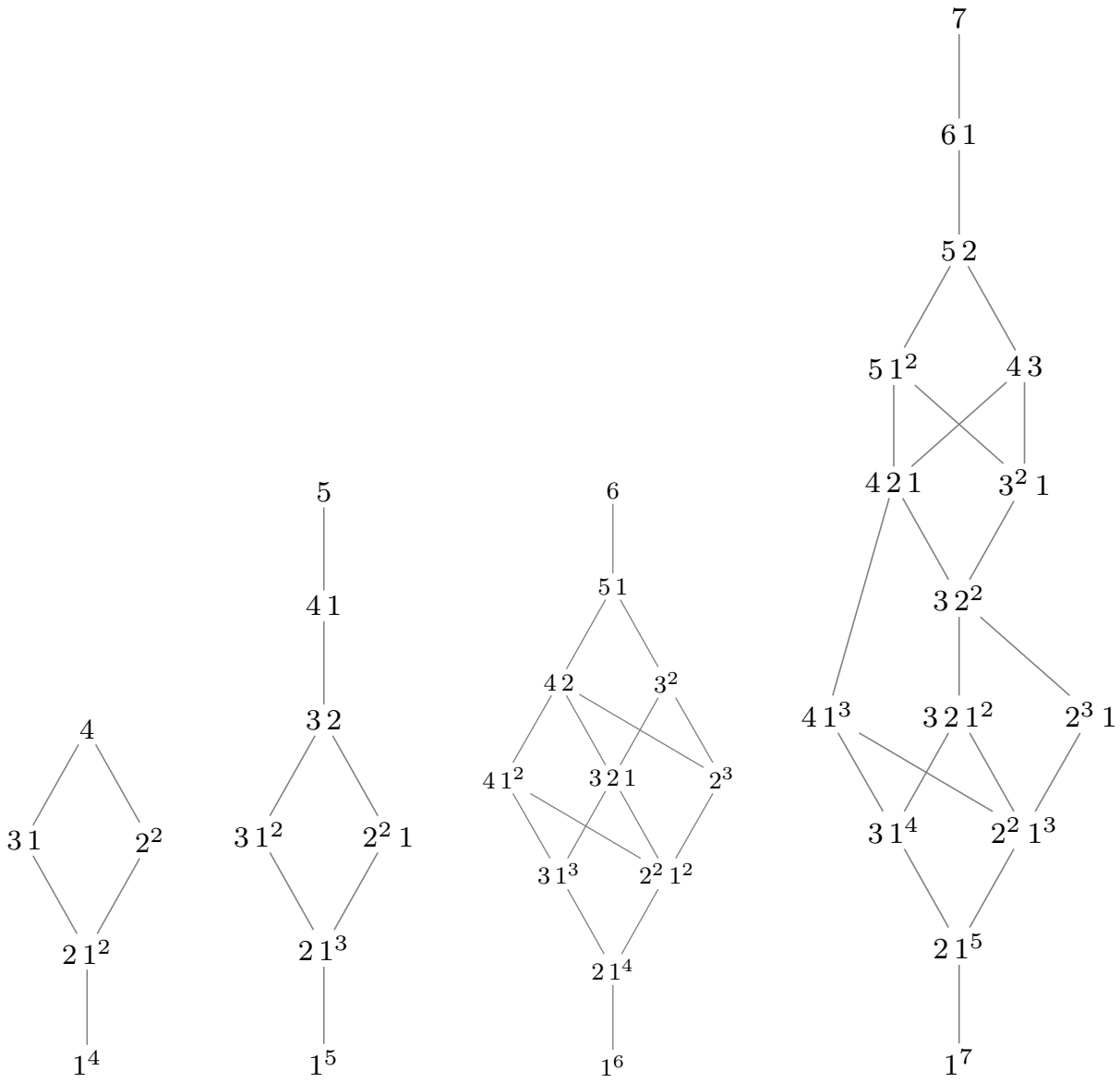


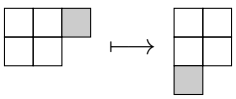
FIGURE 3. The order $\succeq$ of normalised immanants on TNN_n for $n = 4, 5, 6, 7$. Elements are placed at the height of the longest chain below them.

8.4. **Pate's inequalities.**

Corollary 8.3. *Let $n \leq 10$, let $\lambda \vdash n$ with $\lambda_1 \geq 2$, and let μ be obtained from λ by removing its last column and adding its boxes to the bottom of the first column. Then $N_\lambda \geq N_\mu$ on TNN_n, and the difference is KL-positive.*

Proof. Pate [16] proved $N_\lambda \geq N_\mu$ on positive semidefinite matrices, so the difference lies in C_J, which equals C_{KL} for $n \leq 10$ by Corollary 1.4. □

Example 8.B. For $\lambda = (3, 2)$ the last column has one box, which moves to the bottom of the first column, giving $\mu = (2, 2, 1)$:



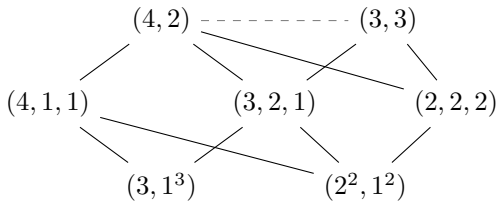


FIGURE 4. Part of the order $\succeq$ for $n = 6$ (Theorem 8.2(2)). Solid lines are covers of $\succeq$ among the seven partitions shown. In dominance $(4,2)$ is above $(3,3)$ (dashed), but in $\succeq$ they are incomparable. Their common lower bounds have two maximal elements, $(3,2,1)$ and $(2,2,2)$, which are incomparable, so $(4,2) \wedge (3,3)$ does not exist. $(4,1,1)$ lies below $(4,2)$ but not below $(3,3)$.

Here $f^\lambda = f^\mu = 5$, and for $\nu = (5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1^3), (1^5)$,

$$K_{\lambda,\nu} = (0,0,1,1,2,3,5), \qquad K_{\mu,\nu} = (0,0,0,0,1,2,5),$$

so $K_{\lambda,\nu}/f^\lambda - K_{\mu,\nu}/f^\mu = \frac{1}{5}(0,0,1,1,1,1,0) \geq 0$, as Theorem 1.5 requires.

Independently of Pate's theorem, the Kostka criterion of Theorem 1.5 holds for every Pate pair with $n \leq 12$.

8.5. **An inequality between several normalised immanants.** At the identity matrix every N_a equals 1, so the permanent cannot dominate a sum of two or more of them. Measured against the determinant, however, it can.

Proposition 8.4. *Let $n \leq 10$ and let S be a set of partitions of n different from (n) and (1^n). Then*

$$\mathrm{per} - \det \;\geq\; \sum_{a \in S} \big(N_a - \det\big) \quad \text{on } \mathrm{TNN}_n \qquad \Longleftrightarrow \qquad \sum_{a \in S} \frac{K_{a,\mu}}{f^a} \leq 1 \quad \text{for all } \mu \neq 1^n.$$

Proof. The difference of the two sides is d_ϕ with $\phi = \chi^{(n)} + (|S|-1)\chi^{(1^n)} - \sum_{a \in S} \chi^a/f^a$. We have $\langle \phi, \eta^\mu \rangle = 1 - \sum_S K_{a,\mu}/f^a$ for $\mu \neq 1^n$, and $\langle \phi, \eta^{1^n} \rangle = 0$. Apply Corollary 1.4. □

The largest such sets have $1,2,2,3,3,4,4,4$ elements for $n = 3, \ldots, 10$. The smallest example in which total non-negativity is essential occurs at $n = 6$.

Example 8.C. On TNN_6,

$$\mathrm{per} + 2\det \;\geq\; \frac{\mathrm{Imm}_{\chi^{3111}}}{10} + \frac{\mathrm{Imm}_{\chi^{222}}}{5} + \frac{\mathrm{Imm}_{\chi^{21111}}}{5}. \tag{5}$$

After multiplying by 10, the values of $(\text{left} - \text{right})(J_\mu)/\mu!$ are:

μ	6	51	42	411	33	321	31^3	2^3	$2^2 1^2$	21^4	1^6
value	10	10	10	10	10	10	9	8	7	0	0

Equality holds at J_{21^4} and at the identity.

Proposition 8.5. *Inequality (5) fails on some entrywise non-negative matrices, and it is not a non-negative combination of two-term inequalities $\alpha N_a - \beta N_b \geq 0$ valid on TNN_6 ($\alpha, \beta \geq 0$). This includes $\mathrm{per} \geq N_a$, $N_a \geq 0$, and every relation of $\succeq$.*

Proof. For the permutation matrix of a 6-cycle, ten times the difference of the two sides is the value of the class function at a 6-cycle, which is -11. Suppose $T = \mathrm{per} + 2\det - N_{3111} - N_{222} - N_{21111}$ is a combination $\sum_i (\alpha_i N_{a_i} - \beta_i N_{b_i})$ of such inequalities. Each term is non-negative at the identity,

so $\alpha_i \ge \beta_i$. Since $T(I) = 0$, every term has $\alpha_i = \beta_i$. The N_a are linearly independent, so the coefficient 2 of $N_{1^6} = \det$ must come from a term $\alpha_i(\det - N_b)$ with $b \ne 1^6$. But $\det(J_b) = 0 < N_b(J_b)$, so such a term is not valid on TNN_6. □

9. Beyond class functions

Let Π_n be the $n!$-dimensional span of the permutation monomials $x^v = x_{1,v(1)} \cdots x_{n,v(n)}$. Every $F \in \Pi_n$ is uniquely $F = \sum_w a_w \operatorname{Imm}_w$, and we put

$$C_{\mathrm{KL}}(\Pi_n) = \{F : \ a_w \ge 0 \ \forall w\}, \qquad C_{\mathrm{TNN}}(\Pi_n) = \{F : \ F \ge 0 \text{ on } \mathrm{TNN}_n\}.$$

Then $C_{\mathrm{KL}}(\Pi_n) \subseteq C_{\mathrm{TNN}}(\Pi_n)$, and the class-function cones are their intersections with the immanants d_ϕ. Let $E \subseteq \mathbb{R}^{S_n}$ be the closed convex cone generated by the evaluation vectors $(\operatorname{Imm}_v(A))_{v \in S_n}$, $A \in \mathrm{TNN}_n$. It lies in the non-negative orthant, and $C_{\mathrm{TNN}}(\Pi_n)$ is its dual cone. Call w *TNN-isolated* if the coordinate vector e_w lies in E. By separation,

$$w \text{ is TNN-isolated} \iff a_w \ge 0 \text{ for every } F = \sum_v a_v \operatorname{Imm}_v \in C_{\mathrm{TNN}}(\Pi_n),$$

so $C_{\mathrm{KL}}(\Pi_n) = C_{\mathrm{TNN}}(\Pi_n)$ iff every w is TNN-isolated. These cones go back to Stembridge [25, §5], who studied the cone of total positivity in $\mathbb{R}S_n$, whose dual is $C_{\mathrm{TNN}}(\Pi_n)$, and to Rhoades–Skandera [19, §7]. Rhoades and Skandera noted $C_{\mathrm{KL}}(\Pi_n) \subsetneq C_{\mathrm{TNN}}(\Pi_n)$ for $n \ge 4$ and asked whether every smooth w is TNN-isolated [19, Question 7.1]. Skandera [21, Thms. 5.4, 5.5] proved that it is, for every n: he constructs $A_w \in \mathrm{TNN}_n$ with $\operatorname{Imm}_v(A_w) = \delta_{v,w}$.

Charts. A *chart* is a sequence of elementary factors $I + tE_{i+1,i}$ and $I + tE_{i,i+1}$ whose lower letters form a reduced word of w_0, as do its upper letters, in any interleaving. By Fomin–Zelevinsky [7], positive parameters give every totally positive matrix up to a positive diagonal factor, which multiplies every element of Π_n by its determinant. Since totally positive matrices are dense in TNN_n, an F whose expansion in some chart has only non-negative coefficients is non-negative on TNN_n. In a fixed chart, $\operatorname{Imm}_w = \sum_\alpha \kappa_\alpha(w) t^\alpha$ with $\kappa_\alpha(w) \ge 0$. Write $\kappa_\alpha = (\kappa_\alpha(w))_w$, and let $\mathcal{N}$ be the Newton polytope, the convex hull of the exponents α with $\kappa_\alpha \ne 0$.

Theorem 9.1 (isolation criterion)**.** *w is TNN-isolated iff some vertex α of $\mathcal{N}$ has* $\operatorname{supp} \kappa_\alpha = \{w\}$.

Proof. Suppose α is such a vertex, and let c be a linear functional maximised on $\mathcal{N}$ only at α. Put $t = s^c$ and let $s \to \infty$. The evaluation vector, divided by $s^{c \cdot \alpha}$, tends to κ_α, which is a positive multiple of e_w. So $e_w \in E$.

Conversely, suppose $e_w \in E$. Write $\mathrm{ev}(A) = (\operatorname{Imm}_v(A))_{v \in S_n}$ and, when $\mathrm{ev}(A) \ne 0$, $u(A) = \mathrm{ev}(A)/\|\mathrm{ev}(A)\|_1$; these normalised vectors generate E as a closed cone. We show in four steps that e_w is a limit of $u(A_k)$ for matrices A_k in the fixed chart, and then read off the vertex.

Step 1: e_w is a limit of normalised evaluation vectors. Since e_w spans an extreme ray of the non-negative orthant, which contains E, this holds: if $\sum_k c_k u_k \to e_w$ with $c_k \ge 0$ and all $u_k \ge 0$ of unit ℓ^1-norm, then $\sum_k c_k \to 1$ and $\sum_k c_k(1 - u_k(w)) \to 0$, so some u_k have $u_k(w) \to 1$, hence $u_k \to e_w$.

Step 2: totally positive matrices suffice. The map $A \mapsto u(A)$ is continuous where $\mathrm{ev}(A) \ne 0$, and totally positive matrices are dense in TNN_n. So every $u(A)$, $A \in \mathrm{TNN}_n$, is a limit of $u(A')$ with A' totally positive, and e_w is a limit of $u(A_k)$ with all A_k totally positive.

Step 3: one fixed chart. By Fomin–Zelevinsky [7], for the fixed chart every totally positive matrix is $D\,M(t)$, where $M(t)$ is the product of the elementary factors of the chart at positive parameters t and D is a positive diagonal matrix. A diagonal factor in any position can be moved to the left, because $D(I + tE_{ij}) = (I + td_i d_j^{-1} E_{ij})D$ only rescales a parameter by a positive number. Every $F \in \Pi_n$ satisfies $F(DM) = \det D \cdot F(M)$, so $u(D\,M(t)) = u(M(t)) = \kappa\tau(t)/\|\kappa\tau(t)\|_1$, where $\mathcal{A}$ is the set of exponents α with $\kappa_\alpha \ne 0$, κ is the matrix with columns κ_α, and $\tau(t) = (t^\alpha)_{\alpha \in \mathcal{A}}/\|(t^\alpha)_\alpha\|_1$.

Step 4: the toric closure. As t ranges over the positive orthant, $\tau(t)$ ranges over the positive part of the projective toric variety of $\mathcal{A}$. Its closure is the disjoint union, over the faces $\mathcal{F}$ of $\mathcal{N}$, of the positive parts of the torus orbits, whose points y satisfy $y_\alpha > 0$ for $\alpha \in \mathcal{F}$ and $y_\alpha = 0$ otherwise [23]. On this compact set the map $y \mapsto \kappa y/\|\kappa y\|_1$ is continuous, because $\kappa \geq 0$ has no zero column, so $\|\kappa y\|_1 \geq \min_\alpha \|\kappa_\alpha\|_1 > 0$ for $y \geq 0$ with $\|y\|_1 = 1$. So the limit e_w equals $\kappa y/\|\kappa y\|_1$ for some y in the closure, supported exactly on some face $\mathcal{F}$, that is, e_w is a positive multiple of $\sum_{\alpha\in\mathcal{F}} \kappa_\alpha y_\alpha$ with all $y_\alpha > 0$. Since all $\kappa_\alpha \geq 0$, every κ_α with $\alpha \in \mathcal{F}$ is supported on $\{w\}$. In particular this holds for a vertex of $\mathcal{F}$, which is a vertex of $\mathcal{N}$. □

Theorem 9.2. (1) *For $n \leq 3$ every $w \in S_n$ is TNN-isolated, so $C_{\mathrm{KL}}(\Pi_n) = C_{\mathrm{TNN}}(\Pi_n)$.*

(2) *For $n = 4$ and $n = 5$, w is TNN-isolated iff w avoids* 3412 *and* 4231. *These are* 22 *of the* 24 *elements of S_4 and* 88 *of the* 120 *elements of S_5. In particular $C_{\mathrm{KL}}(\Pi_n) \subsetneq C_{\mathrm{TNN}}(\Pi_n)$ for $n = 4, 5$.*

Both directions of part (2) are proved here directly from Theorem 9.1, without using [21] or [13]. For every n, "smooth ⇒ isolated" is due to Skandera [21], and the companion paper gives a proof of the converse by a different method [13, Thm. 9.6].

Proof. Apply Theorem 9.1 to the chart whose lower and upper words are both $s_1\, s_2 s_1\, s_3 s_2 s_1 \cdots s_{n-1} \cdots s_1$, all lower factors first, by computing the vertices of $\mathcal{N}$ that carry a single κ-support. All certificates are exact and archived (`results/tnn_isolation_certificates_n4.json` and `_n5.json`). For each isolated w there is an exponent α with $\operatorname{supp}\kappa_\alpha = \{w\}$ and an integer functional maximised on $\mathcal{N}$ only at α. For each singular w, every exponent α with $\operatorname{supp}\kappa_\alpha = \{w\}$ is the midpoint of two other exponents, hence not a vertex (1 and 18 such exponents for 3412 and 4231; 4,747 in all for $n = 5$, and some singular w have none). Both kinds of certificate are re-checked in integer arithmetic by `check_isolation_certificates.py --verify`. □

Example 9.A. For $n = 3$ the chart of the proof is $L_1(t_1)L_2(t_2)L_1(t_3)\,U_1(u_1)U_2(u_2)U_1(u_3)$, with $L_i(t) = I + tE_{i+1,i}$ and $U_i(t) = I + tE_{i,i+1}$. The six Imm_w have 14 distinct exponents in total. For each w the table gives a monomial t^α that occurs in Imm_w with coefficient 1 and in no other Imm_v, and a functional c with $c\cdot\alpha > c\cdot\beta$ for the other 13 exponents β. So α is a vertex of $\mathcal{N}$ with $\kappa_\alpha = e_w$, and Theorem 9.1 shows that every $w \in S_3$ is TNN-isolated.

w	t^α	c on $(t_1, t_2, t_3, u_1, u_2, u_3)$
123	1	$(-1,-1,-1,-1,-1,-1)$
132	t_2u_2	$(-1,1,-1,-1,1,-1)$
213	t_3u_3	$(-1,-1,1,-1,-1,1)$
231	$t_2t_3u_2u_3$	$(-1,1,0,-1,1,1)$
312	$t_1t_2u_1u_2$	$(1,1,-1,0,1,-1)$
321	$t_2t_3^2u_1u_2u_3$	$(-1,1,1,0,1,1)$

Explicit elements of $C_{\mathrm{TNN}}(\Pi_5) \setminus C_{\mathrm{KL}}(\Pi_5)$ are given in [13, Thm. 9.2], for example $\mathrm{Imm}_{21543} + \mathrm{Imm}_{54321} - \mathrm{Imm}_{52431}$. For $n = 4$, Skandera and Zelevinsky give $\mathrm{Imm}_{3214} + \mathrm{Imm}_{1432} - \mathrm{Imm}_{3412} \in C_{\mathrm{TNN}}(\Pi_4) \setminus C_{\mathrm{KL}}(\Pi_4)$ (reported in [21, (5.1)]), certified by a subtraction-free expression in minors. The stronger element $\frac{1}{2}(\mathrm{Imm}_{1432} + \mathrm{Imm}_{3214}) - \mathrm{Imm}_{3412}$ is non-negative on TNN_4 by [13, Prop. 9.3]. Lifting it, and the elements $\varepsilon\,\mathrm{Imm}_{2143} + \frac{3}{4\varepsilon}\,\mathrm{Imm}_{4321} - \mathrm{Imm}_{4231}$ for small $\varepsilon > 0$, along an occurrence of 3412 or 4231 gives a proof for every n that every TNN-isolated w is smooth [13, Thm. 9.6], the converse of Skandera's theorem. Theorem 9.2 confirms this for $n \leq 5$ by a different method.

10. Further computations

This section records computations about results that are proved, or conjectured, elsewhere, mostly in the companion preprint [13]. They are cross-checks of those proofs and evidence for those conjectures, not replacements for them, and nothing in Sections 1–9 depends on them.

10.1. **Clean steps.** A step $w \to ws$ with $sw < w < ws$ is *clean* if no $z < w$ with $zs < z$ has $\mu(z, w) \neq 0$; then $c_{\cdot,ws} = 2c_{\cdot,w}$ [13, §7.2]. The steps certified by the bad-cover criterion of [13, Prop. 7.8], starting from smooth permutations, reach 84%, 68%, 49%, 32% and 19% of the singular permutations for $n = 5, \ldots, 9$; all clean steps together reach 84%, 68%, 51%, 35% and 22%. The singular w that are not the top of a clean step are a stable 30% of all singular w for $n = 7, 8, 9$.

10.2. **Checks of the results for all n.** The results of [13] were also checked by computer. The product theorem [13, Thm. 3.1] holds for all I, T, u, v with $n \leq 6$, and its single-entry case for $n \leq 7$. The identity $\operatorname{ch} D_k = \sum_j h_j m_{(k-j,1^{n-k})}$ of Skandera's hook chain holds for $n \leq 12$, and D_k and every term have non-negative Kazhdan–Lusztig coefficients for $n \leq 8$. For the hook theorem, the closure identity was checked for $m \leq 7$, the formulas for the rotations and for $H_c(e)$ for $m \leq 9$, the boundary signs for $m \leq 10$, and the whole recursion numerically for $m \leq 7$. The diagram structure in the bottom-terms lemma was checked exactly from the Kazhdan–Lusztig polynomials for $m \leq 6$, without the colouring rule.

For the isolation theorem [13, Thm. 9.6] we used the exact values $P_{\sigma,w}(1)$ from the extremal-pair tables. For each singular w with $n \leq 10$ (there are 3,504,490 for $n = 10$), take the first occurrence of 3412 in lexicographic order, or of 4231 if there is none. Lift $\operatorname{Imm}_{1432} + \operatorname{Imm}_{3214} - 2\operatorname{Imm}_{3412}$, respectively $\operatorname{Imm}_{2143} + \operatorname{Imm}_{4321} - \operatorname{Imm}_{4231}$, along it as in the proof. In every case the coefficient of Imm_w is negative. The proof itself needs no choice of occurrence. Its smoothness lemma [13, Lemma 9.5] says that X_w is smooth at $w(i\,j)$ iff the entries of w inside the rectangle spanned by positions i, j decrease, and that then $P_{w(i\,j),w} = 1$. We checked that $P_{w(i\,j),w} = 1$ iff these entries decrease, for every inversion of every w with $n \leq 10$ (81,648,000 inversions for $n = 10$).

10.3. **Hook expansions.** Let $B_{ij}(w)$ be the coefficient of Imm_w in $x_{ij}x_{ji} \det x[\{i,j\}^c, \{i,j\}^c]$. The conjectural closed form for B_{ij} in [13, Conj. 10.1] matches the computation for every pair (i, j) with $n \leq 8$, and the inertness property stated after it holds for every 2- and 3-cycle with $n \leq 8$. For $a \geq 2$, $f_i^{(a)}(C'_w(1)) = 0$ unless w moves a single block of consecutive letters containing i, for $n \leq 8$. The functions g_T in the decomposition of Skandera's θ_ℓ are KL-positive for $n \leq 8$ (`results/theta_per_set_n8.log`).

10.4. **The shapes $(3, 2^b, 1^c)$.** For these shapes Conjecture 1.2 reduces to 321-avoiding w containing both 312 and 231 [13, Props. 7.2, 8.2]. There are 40, 130 and 427 indecomposable such w for $n = 6, 7, 8$, for example for $(3, 2, 1)$, $(3, 2, 1, 1)$ and $(3, 2, 2, 1)$, and Theorem 1.3 covers $3 + 2b + c \leq 10$.

Open problems.

(1) *Scalable and certifiable computation.* The extremal-pair method applies unchanged to $n = 11$, but S_{11} has an estimated 5×10^{10} extremal pairs; extrapolating from $n = 9, 10$, the graded step needs a few times 10^4 core-hours and several terabytes of working memory. Find an algorithm that is sharded by descent classes and streamed to disk, and that emits certificates which can be checked independently and more cheaply than the computation itself, so that a result for $n = 11$ would not rest on a single run.

(2) *The Kostka order.* The criterion $K_{a,\mu}/f^a \geq K_{b,\mu}/f^b$ for all $\mu \vdash n$ of Theorem 1.5 defines a partial order on partitions of every n, contained in dominance order; it agrees with $\succeq$ on TNN_n for $n \leq 10$, and its failures are failures of $\succeq$ for every n. Describe this order

combinatorially: which pairs comparable in dominance are incomparable in it, and what obstructs the lattice property (Theorem 8.2).

(3) *Chart certificates.* For each singular w, an element of $C_{\mathrm{TNN}}(\Pi_n)$ with a negative coefficient at Imm_w and non-negative coefficients elsewhere exists by the proof given in [13, Thm. 9.6]. Find such elements whose expansion in a single chart of Theorem 9.1 is subtraction-free. Our search found them for only 3 of the 32 singular $w \in S_5$ in 200 random charts, and for neither singular $w \in S_4$ in 30 charts (`results/tnn_isolation_n4.json`, `_n5.json`). Decide whether they exist for 3412 and 4231, and whether they can be constructed uniformly, for example by lifting along pattern occurrences. Relatedly, find compact, human-readable Newton-polytope certificates for Theorem 9.2, and use them to give an independent check of the isolation theorem for $n = 6$.

(4) *The graded cones.* For $n \leq 10$, $C_J(q) \subseteq C_{\mathrm{KL}}(q) \subseteq C_J^{\mathrm{ev}}(q)$, the first inclusion strict for $n = 2$ and the second for every $n \geq 3$ (Section 8). Describe $C_{\mathrm{KL}}(q)$, for example by finding its generators as an $\mathbb{N}[q]$-semimodule for $n = 3, 4$, and determine which traces in $C_J^{\mathrm{ev}}(q) \setminus C_J(q)$ lie in it.

Appendix A. Hasse diagrams for $n = 8, 9, 10$

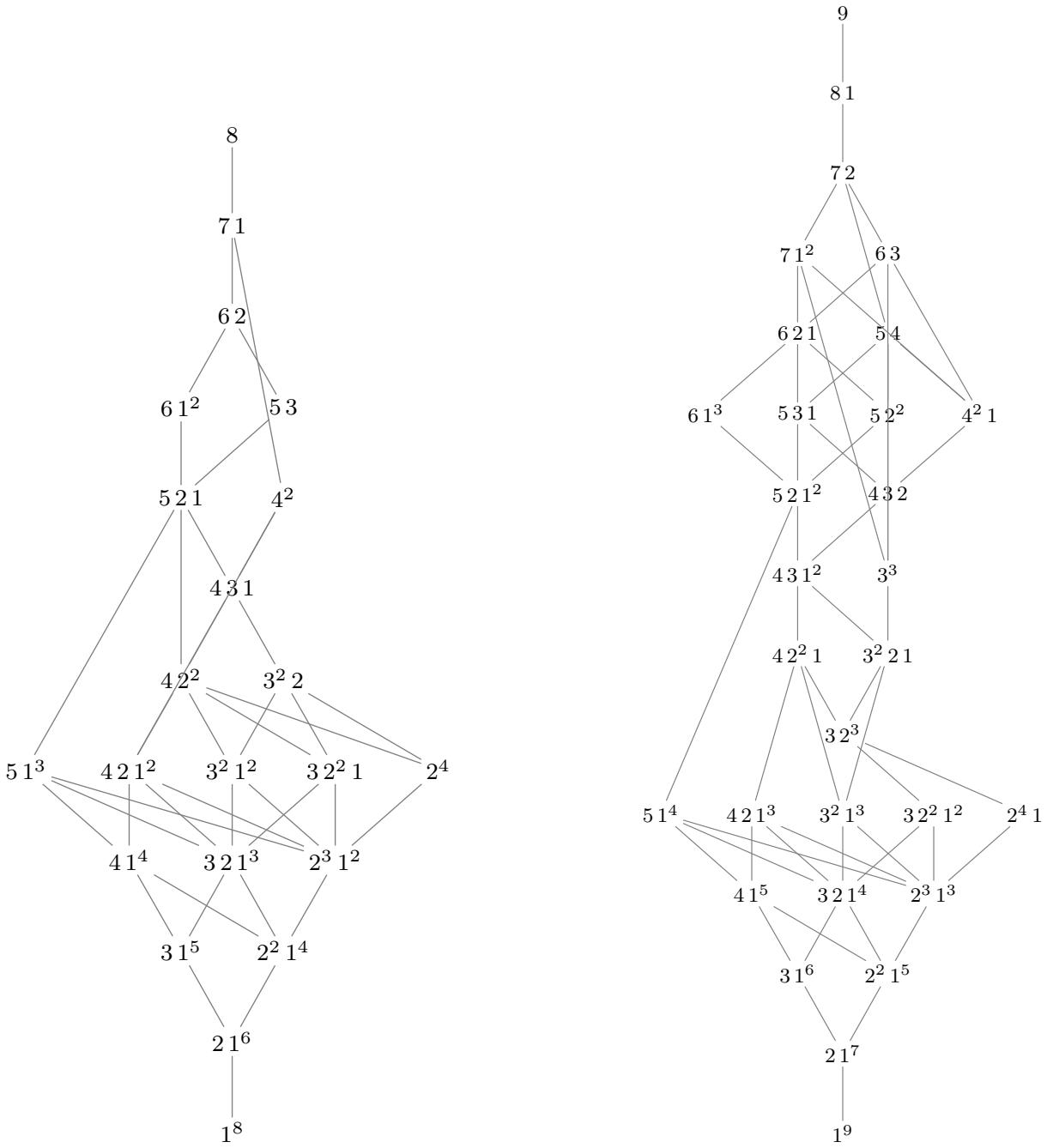


FIGURE 5. The order $\succeq$ of normalised immanants on TNN_8 (left) and TNN_9 (right).

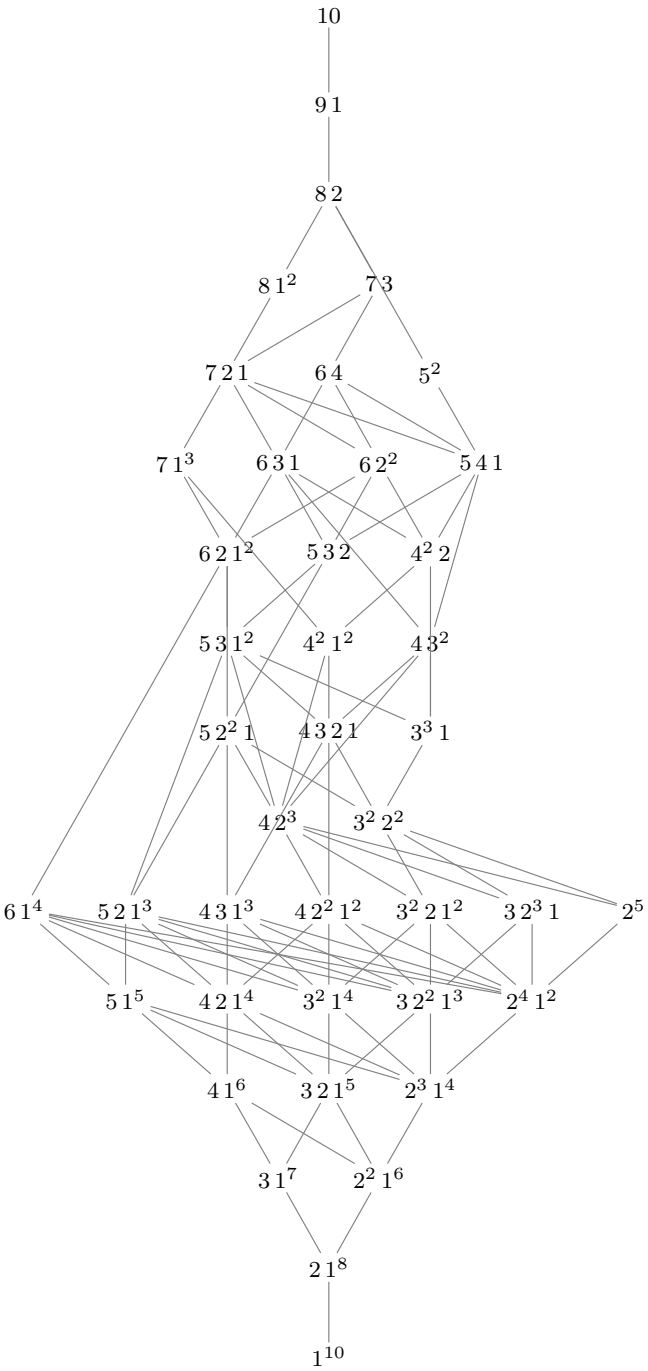


FIGURE 6. The order $\succeq$ of normalised immanants on TNN_{10}.

Appendix B. Code and data

The code and data are in the repository
https://github.com/keithmcnulty/haiman-immanants-papers-n10-hooks; the version used for this paper is the release tagged `v1.4.0`, archived as [14]. The computation of Section 7 and its results are unchanged since the first release, `v1.0.0` (doi:10.5281/zenodo.23435088); `v1.1.0` adds the checks of the rooted refinement [13, Conj. 10.2], `v1.2.0` the $n = 10$ part of check 12, `v1.3.0` its test through all 42 irreducible characters, and `v1.4.0` the check of the g_T for $n = 8$ and the present version of the text. Later releases make only documentation and packaging changes and change no result. All computations used Python with NumPy, Numba, SciPy and python-flint, together with Coxeter3 [4] and SageMath for cross-checks. The repository contains:

- the extremal-pair engine `monimm/kl_extremal.py` and its driver `run_kl_extremal.py`, with the result summary `results/ext_n10.json`;
- the graded engine `monimm/kl_graded.py` and its driver `run_kl_graded.py`, with results `results/kl_graded_n8.json`, `results/kl_graded_n9.json`, `results/kl_graded_n10.json` (cloud runbook `cloud/run_kl_graded_n10.sh`), the induced-module test `tests/test_kl_graded.py`, and the chromatic check `check_codominant_chromatic.py` with `results/kl_graded_n10_codominant.json`;
- the induced-module checks `check_graded_induced.py` ($n = 6$, `results/graded_induced_n6.json`) and `check_induced_traces.py` with the depth-first kernel `monimm/induced_traces.py` (`results/induced_traces_n8.json`, `results/induced_traces_n10.json`); at $n = 10$ it reads the full graded table and the traces written by `cloud/run_kl_graded_n10_tables.sh` (2.5 GB compressed), which are archived with the data record [15];
- the seminormal-form check `check_seminormal_traces.py` with `monimm/seminormal_traces.py` (`results/seminormal_traces_n8.json`, `results/seminormal_traces_n10.json`; cloud runbook `cloud/run_seminormal_n10.sh`, 29 minutes on 192 cores), which writes the table of all $\chi_q^\mu(T_v)$ mod p at $q = 2, 3$, archived with the data record from version 2.1.0 [15];
- the exact inversion check `check_inversion_exact.py`, with results `results/inversion_exact_n9.json` and `results/inversion_exact_n10.json`;
- the Coxeter3 comparison `cross_check_coxeter3.py`, with results `results/coxeter3_check_n*.json`;
- the SageMath comparison of check 9, `cross_check_sage.py`, with results `results/sage_check.json` (every pair of $S_n \times S_n$ for $n \leq 5$, every extremal pair for $n = 6$);
- the full-interval engines `monimm/kl_fast.py` and `monimm/kl_big.py` ($n \leq 9$);
- the order computations `monimm/ordering.py` and `monimm/order_structure.py`, with results `results/immanant_ordering_n1-12.json` and `results/order_structure_n1-10.json`;
- the isolation study `monimm/tnn_cone.py` and `check_tnn_isolation.py`, with results `results/tnn_isolation_n4.json` and `results/tnn_isolation_n5.json`, and the exact certificates for Theorem 9.2, `check_isolation_certificates.py` with `results/tnn_isolation_certificates_n4.json` and `results/tnn_isolation_certificates_n5.json` (exponent vectors, integer separating functionals, and exact non-vertex certificates; `--verify` re-checks a file in integer arithmetic);
- the clean-step computations `notes/clean_steps.py`, with results `results/clean_steps_n*.json`, the checks of [13] in `notes/hook_proof/`, `notes/skandera_chain.py`, `notes/cprod.py`, `notes/r2_entry_check.py`,

`notes/pattern_lifting.py` and, for the rooted refinement [13, Conj. 10.2], `notes/rooted_vertex.py`, `notes/rooted_graded.py` and `notes/rooted_graded_an.py` (logs `results/rooted_*.log`), and the hook expansions and the counts for the shapes $(3, 2^b, 1^c)$ of Section 10 in `check_further_computations.py`, with results `results/further_computations_n8.json`;

- a test suite covering every engine against brute force at small n.

The full tables for $n = 10$ are archived separately as [15] (version 2.1.0, doi:10.5281/zenodo.23532289): the $42 \times 10!$ integers $c_{\lambda,w}$ (`ext_n10.npz`), the $42 \times 10!$ polynomials $c_{\lambda,w}(q)$ (`kl_graded_n10.npz.zst`), the traces $\phi_q^\lambda(T_v)$ modulo $2^{31} - 19$ at $q = 2, 3$ for every v (`kl_graded_n10_traces_modp.npz`), and the irreducible characters $\chi_q^\mu(T_v)$ for every $\mu \vdash 10$ and every v, with the same modulus and the same q (`seminormal_traces_n10_modp.npz`, new in version 2.1.0). Their SHA-256 checksums are in `results/SHA256SUMS`.

Reproducing the results. Each result file has a driver and, for the large runs, a runbook that checks a smaller case first and must reproduce a stored result before the main run starts. The integer-safety bounds of Section 7 are recomputed and recorded by the drivers (`accumulation_bound` and `check_no_int64_overflow` in `results/kl_graded_n10.json`). The command `python -m monimm.verify` runs the checks of Section 7 and prints one line per check, numbered as there: it reruns them for $n \leq 8$ (and $n = 9$ with `--tier local`), checks the stored $n = 10$ results against each other, against the numbers in this paper and against the checksums in `results/SHA256SUMS`, and records the package versions; `requirements-lock.txt` pins them.

result	command (from the repository root)	resources
`results/ext_n10.json`	`bash cloud/run_n10.sh` (runs `run_kl_extremal.py --n 10`; acceptance test $n = 9$ against `results/big_n9.npz`)	279 s, 192 cores, 15.5 GB
`results/kl_graded_n10.json`	`bash cloud/run_kl_graded_n10.sh` (runs `run_kl_graded.py --n 10`; acceptance test $n = 8$)	1 h 31 min, 192 cores, 125 GB
`results/kl_graded_n10_codominant.json`	`bash cloud/run_kl_graded_n10_codominant.sh`	as above
`results/coxeter3_check_n10.json`	`bash cloud/run_coxeter3_n10.sh` (runs `cross_check_coxeter3.py --n 10 --all`)	2.4 h, 190 cores
`results/inversion_exact_n10.json`	`check_inversion_exact.py --n 10 --coeffs results/ext_n10.npz`	about 10 GB
`results/kl_graded_n9.json`	`run_kl_graded.py --n 9`	25 min, 4 cores
`results/tnn_isolation_certificates_n5.json`	`check_isolation_certificates.py --n 5`	minutes, 1 core
test suite	`python -m pytest -q`	minutes
all checks, $n \leq 8$	`python -m monimm.verify`	3 min, 4 cores

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic), through Claude Code, to write and test the software, to run and cross-check the computations, to search and summarise the literature, to check arguments against computed examples, and to draft and edit the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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